scqfig gallery

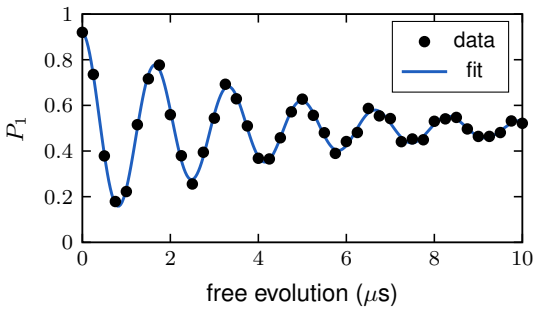
Every example in both styles: **field** (left) and **folio** (right). Build with `python examples/build_gallery.py`; each panel comes from `examples/NN_name.py`.

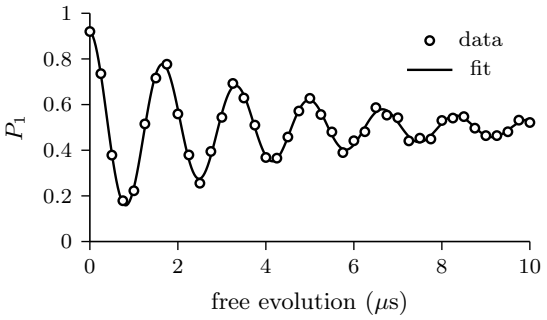
The LaTeX package

`\usepackage[field]{scqfig}` or `[folio]` defines the palette, both styles' roles (`scqFieldAccent`, `scqFolioAccent`) and the active style's roles under plain names (`scqAccent`). `\scqgraphics{t1_decay}` loads the figure of the active style, and the pgfplots styles `scqfig`, `scqfig data`, `scqfig model` and `scqfig ref` match the matplotlib output.

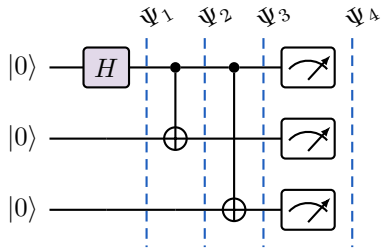
colour	field	folio
scqInk		
scqData		
scqModel		
scqSim		
scqReference		
scqAccent		
scqSignal		
scqBus		
scqWash		
scqOverlay		
scqQubitA		
scqQubitB		
scqQubitC		

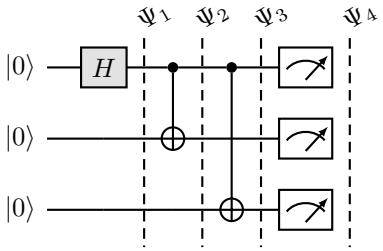
colour	field	folio
scqQubitD		
scqLevel0		
scqLevel1		
scqLevel2		
scqLevel3		
scqTileNonlinear		
scqTileCapacitive		
scqTileInductive		
scqChipPad		
scqChipResonator		
scqChipDrive		
scqChipFlux		





pgfplots axes drawn natively by LaTeX with `[scqfig field]` (left) and `[scqfig folio]` (right).





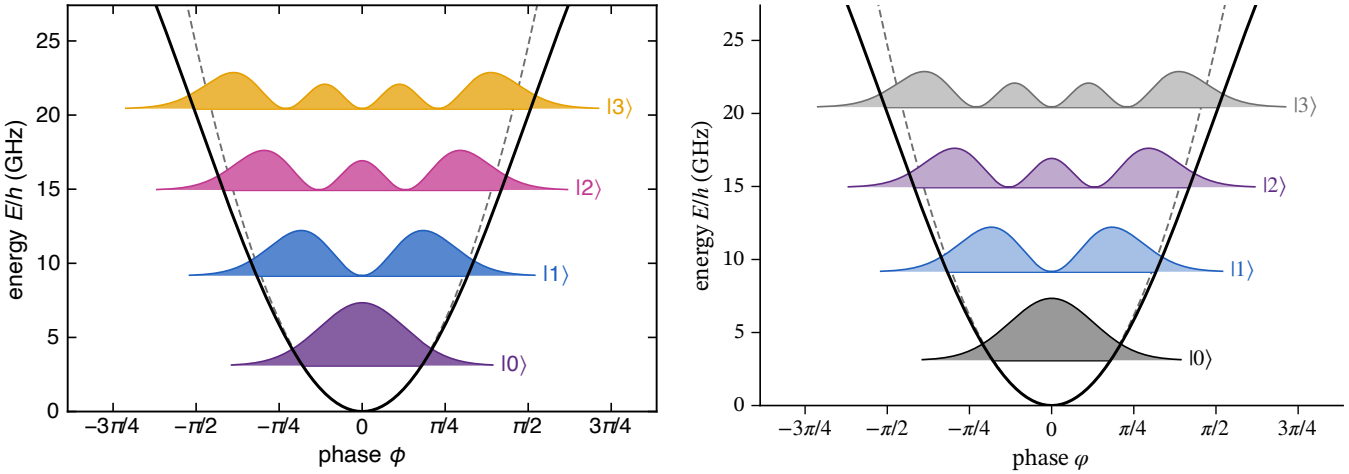
The GHZ circuit of example 78 typeset by quantikz with `[scqfig circuit field]` (left) and `[scqfig circuit folio]` (right). Each slice tilts its label with `\slice[label style=scq slice label]`.

Spectrum

01 Potential well and wavefunctions

`01_potential_well.py`

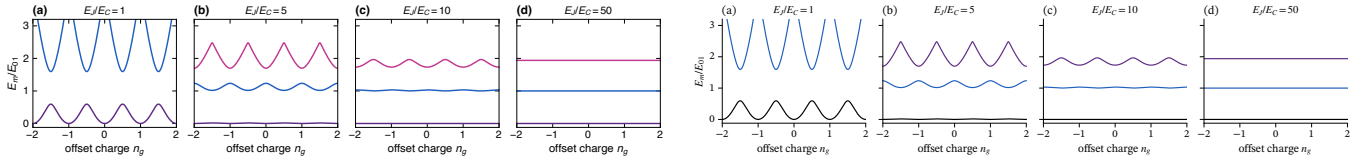
Transmon potential with the four lowest levels and their computed $|\psi(\varphi)|^2$ (charge basis, E_J/E_C = 80).



02 Charge dispersion

`02_charge_dispersion.py`

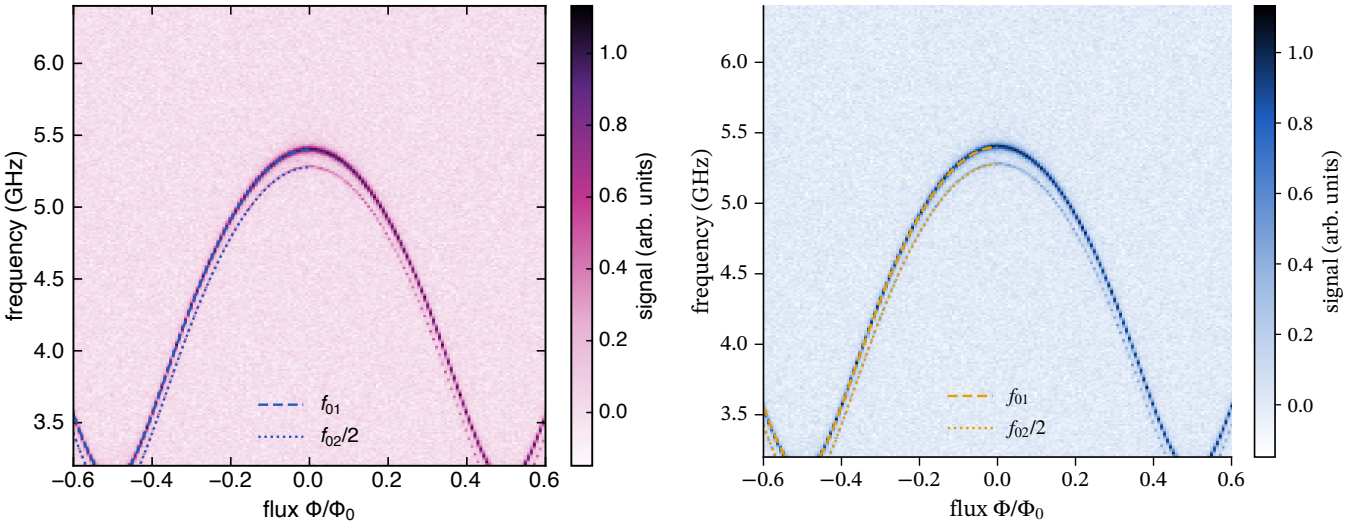
Charge dispersion of the three lowest levels as E_J/E_C grows from the Cooper-pair box to the transmon.



03 Flux-tuned spectroscopy

`03_flux_spectroscopy.py`

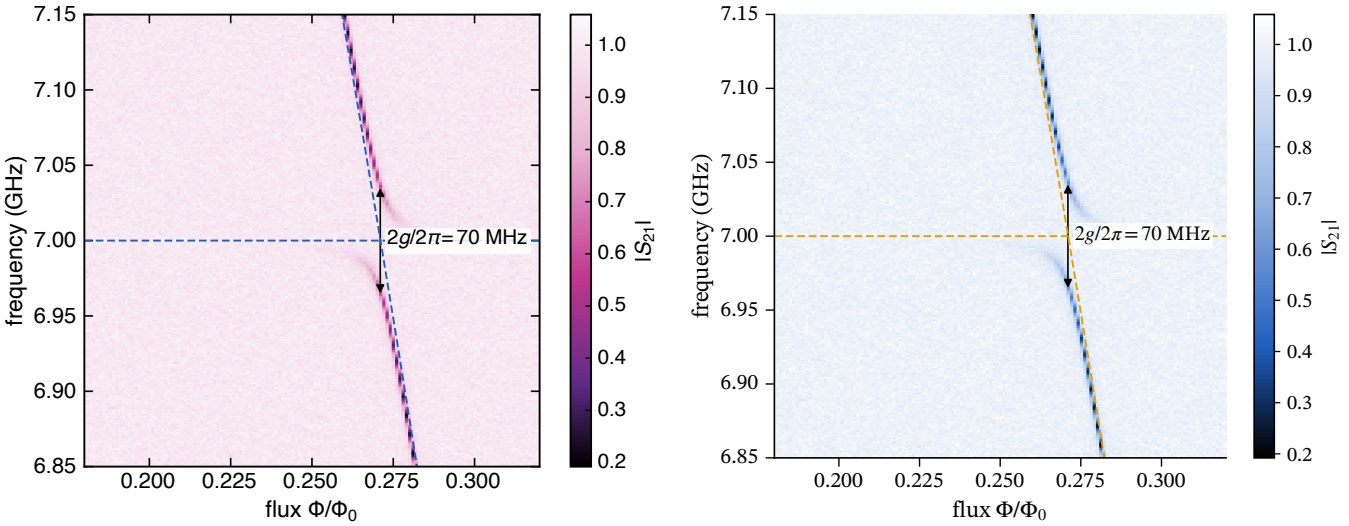
Two-tone spectroscopy of a SQUID transmon against flux, with the exact f01 and f02/2 drawn over the data (synthetic data).



04 Avoided crossing

`04_avoided_crossing.py`

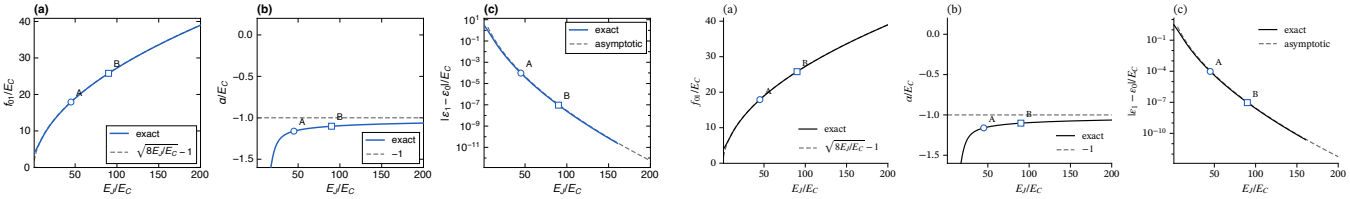
Vacuum-Rabi splitting: resonator transmission as a tunable qubit crosses it, with the bare modes dashed and 2g marked (synthetic data).



05 Scaling with E_J/E_C

`05_ratio_scaling.py`

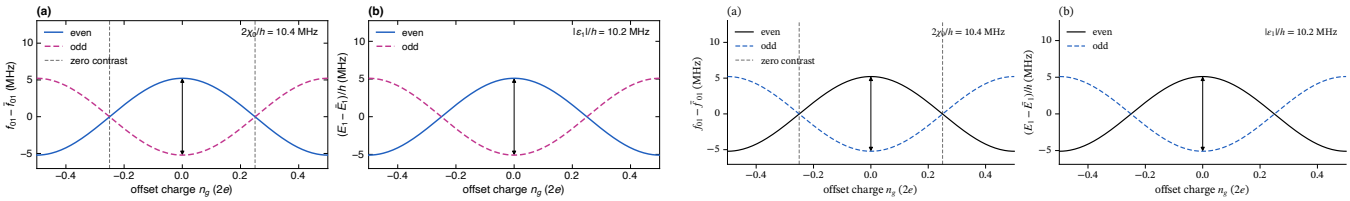
Transmon frequency, anharmonicity and charge dispersion against E_J/E_C, exact in the charge basis with the large-ratio asymptotes dashed and two designs marked (computed).



06 Parity bands

`06_parity_bands.py`

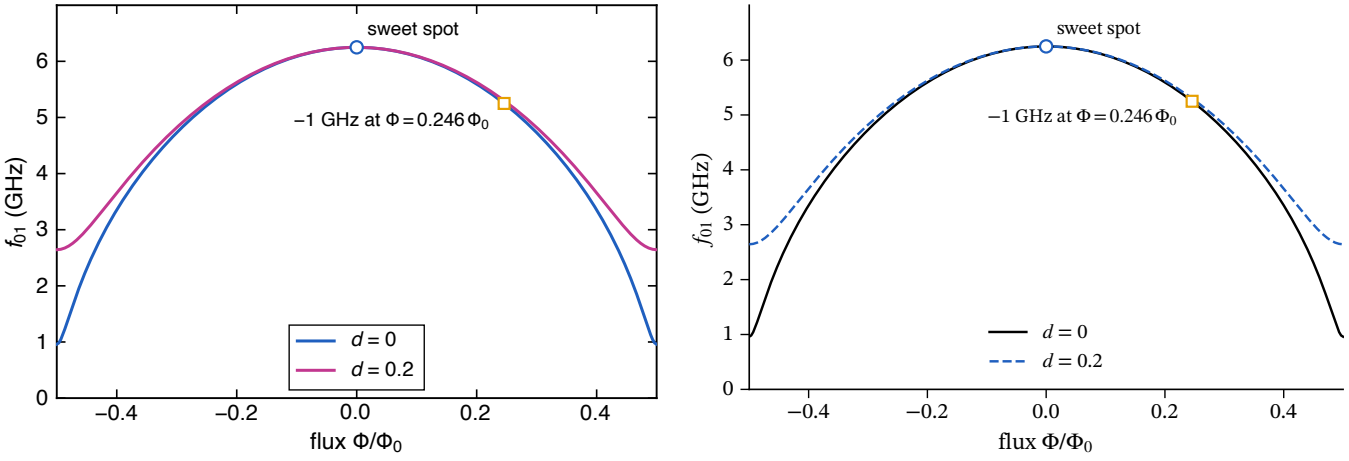
Even and odd charge-parity branches of a charge-sensitive transmon at E_J/E_C = 20: the 0-1 frequency with its splitting 2 chi_0 and the zero-contrast offsets, and the first excited level (computed).



07 Flux tuning

`07_flux_tuning.py`

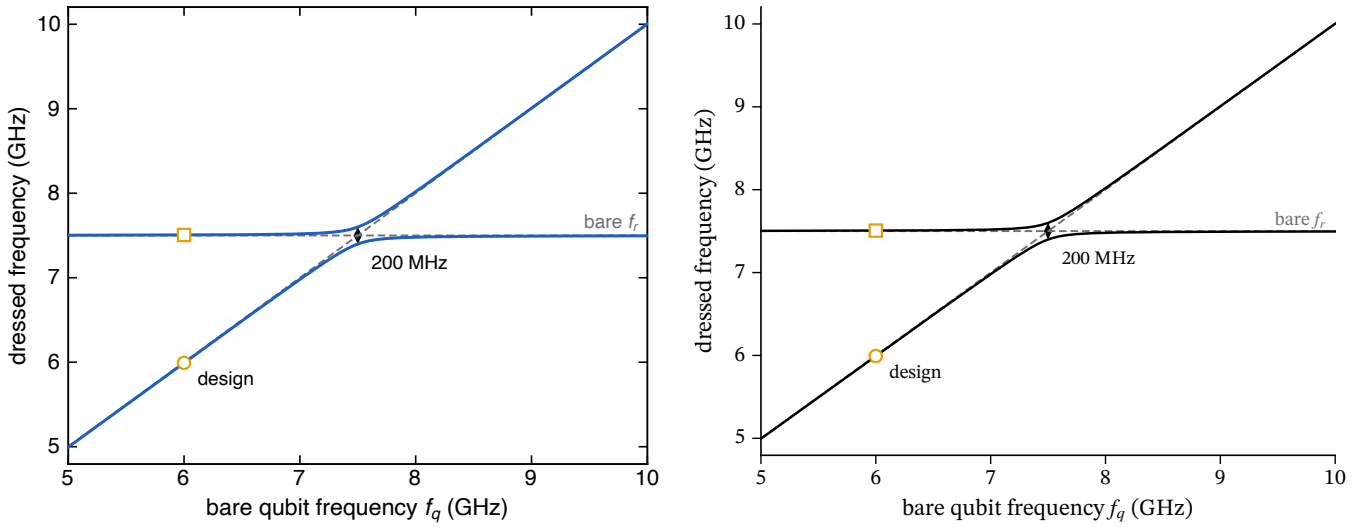
f01 of a SQUID transmon against flux for a symmetric SQUID and a 20 % junction asymmetry, with the sweet spot and the flux that tunes it 1 GHz lower (computed).



08 Dressed lines

`08_dressed_lines.py`

Dressed one-excitation frequencies of a transmon (E_C = 250 MHz) and a 7.5 GHz resonator coupled with g = 100 MHz, swept through the crossing, with the splitting and a design point marked (computed).

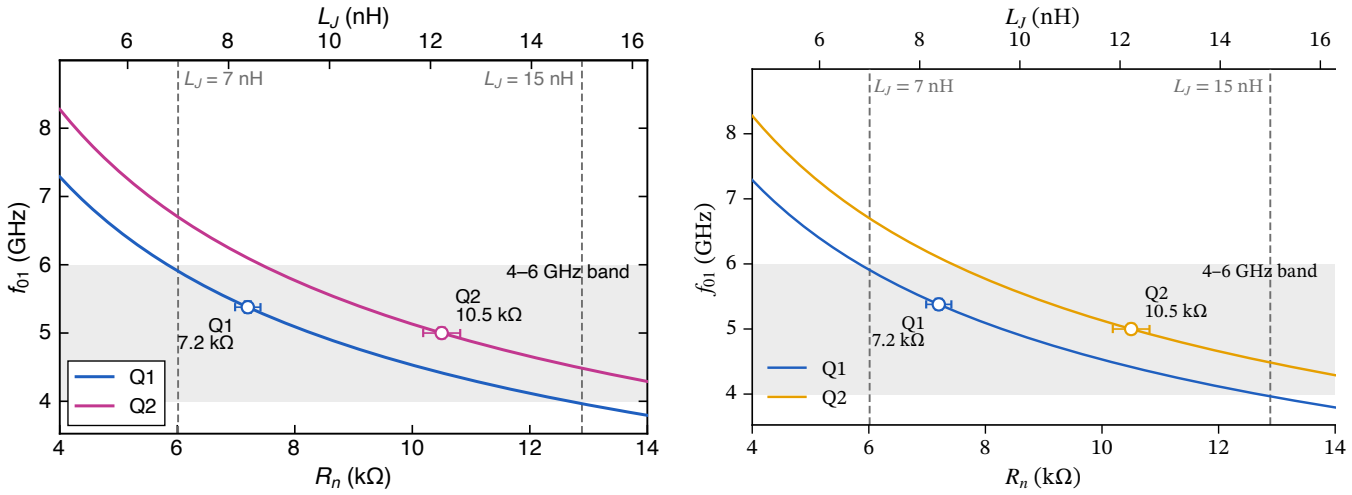


Design

09 Junction design

`09_junction_curve.py`

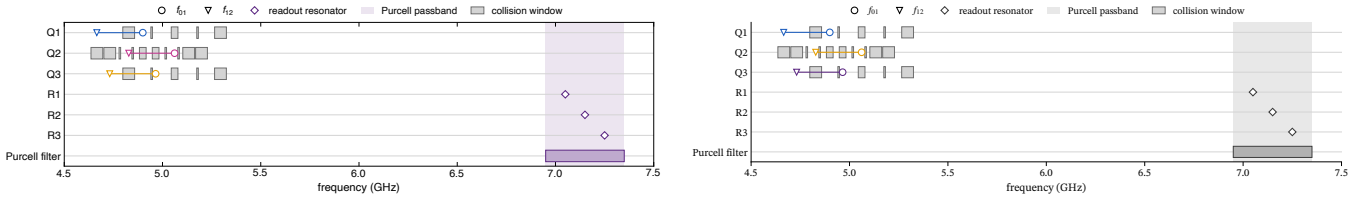
Qubit frequency against junction resistance for two transmons on an aluminium junction (gap 180 μeV), with L_J on the top axis, a 4-6 GHz band, an allowed L_J window and the designs with a 3 % junction scatter (computed).



10 Frequency plan

`10_frequency_plan.py`

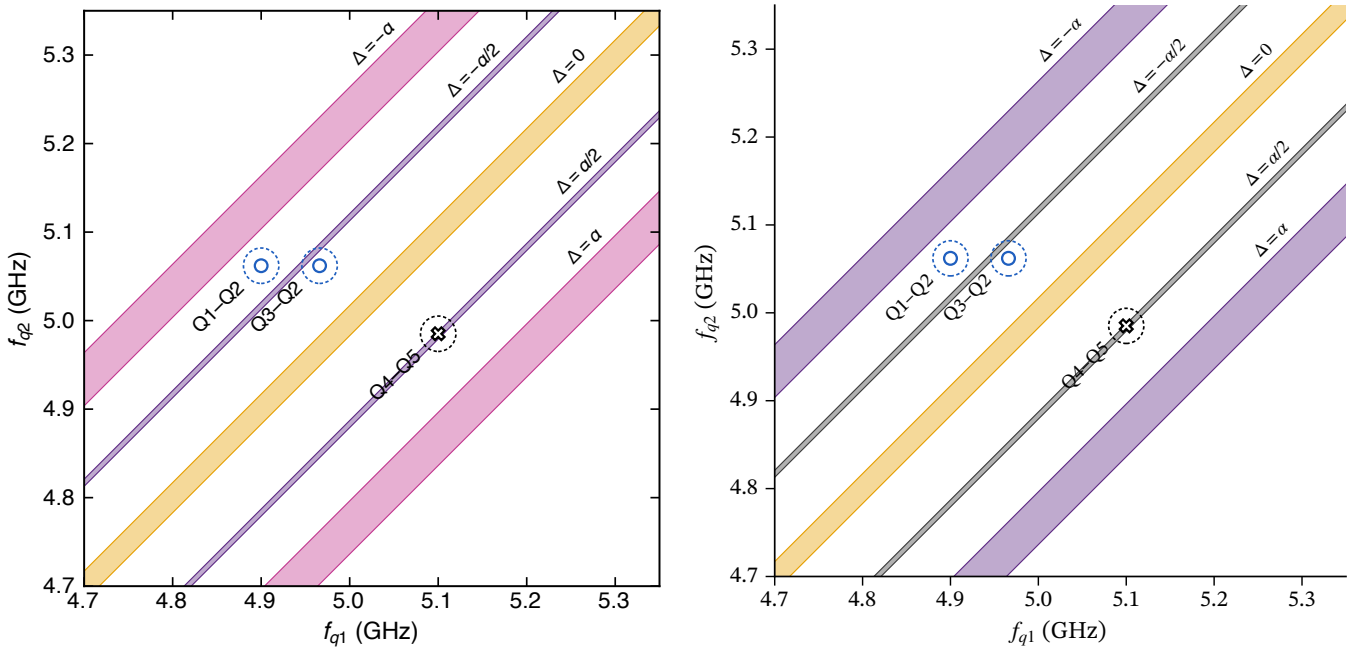
Frequency plan of a three-qubit chain: each transmon's f01 and f12, the readout resonators inside the Purcell-filter passband, and the collision windows each qubit's f01 must avoid (designed values).



11 Collision map

11_collision_map.py

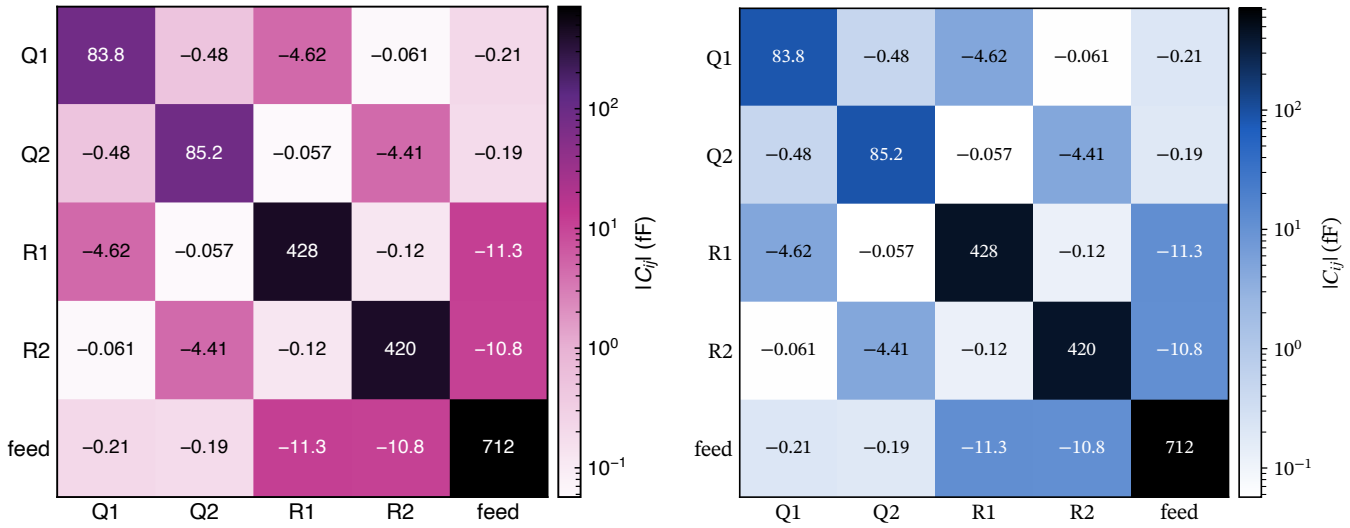
Collision map of two coupled transmons: forbidden bands of the f_q2 against f_q1 plane, with three design points and their 20 MHz fabrication scatter; one point sits on a two-photon band (designed values).



12 Capacitance matrix

12_capacitance_matrix.py

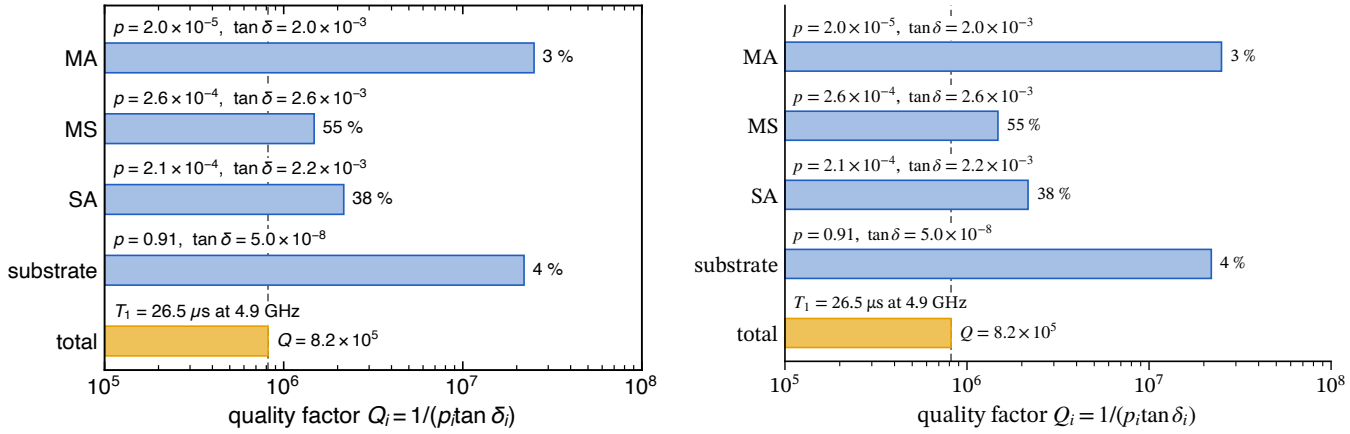
Maxwell capacitance matrix of two transmons, their readout resonators and the feedline, coloured by |C| on a log scale (simulated values).



13 Participation budget

13_participation.py

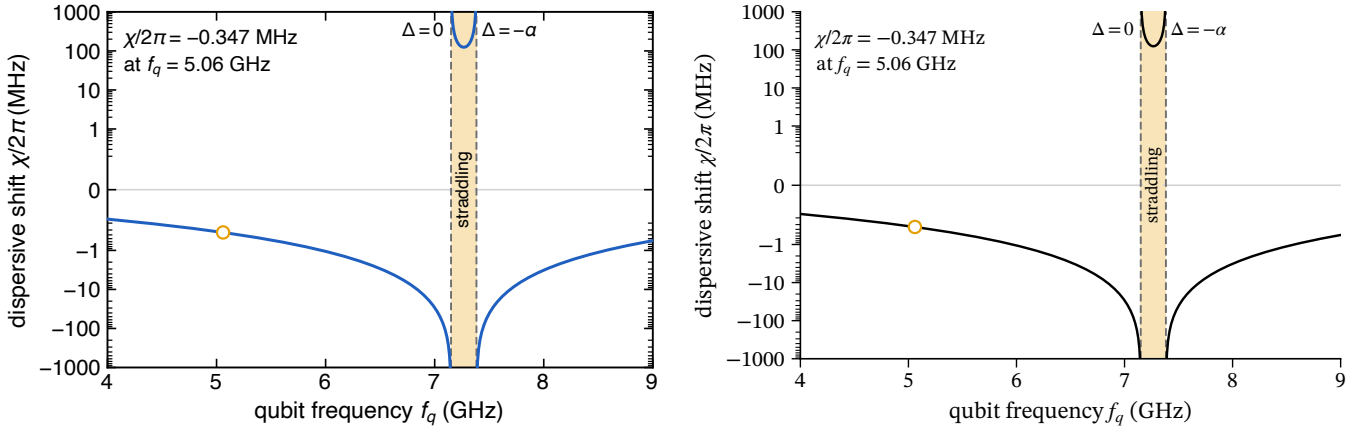
Surface-participation loss budget of a transmon: participation and loss tangent of each region, the quality factor each allows, and the total (simulated participation, assumed loss tangents).



14 Dispersive shift

`14_dispersive_shift.py`

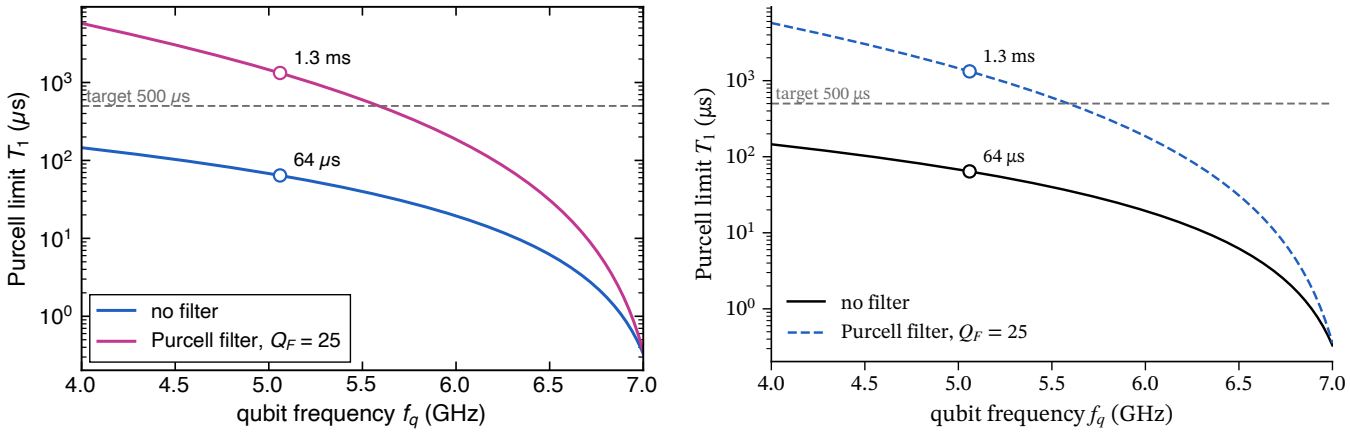
Dispersive shift of a transmon against its frequency for a 7.15 GHz resonator, g = 85 MHz and alpha = -233 MHz, with the straddling regime shaded and the design point marked (computed).



15 Purcell limit

`15_purcell_limit.py`

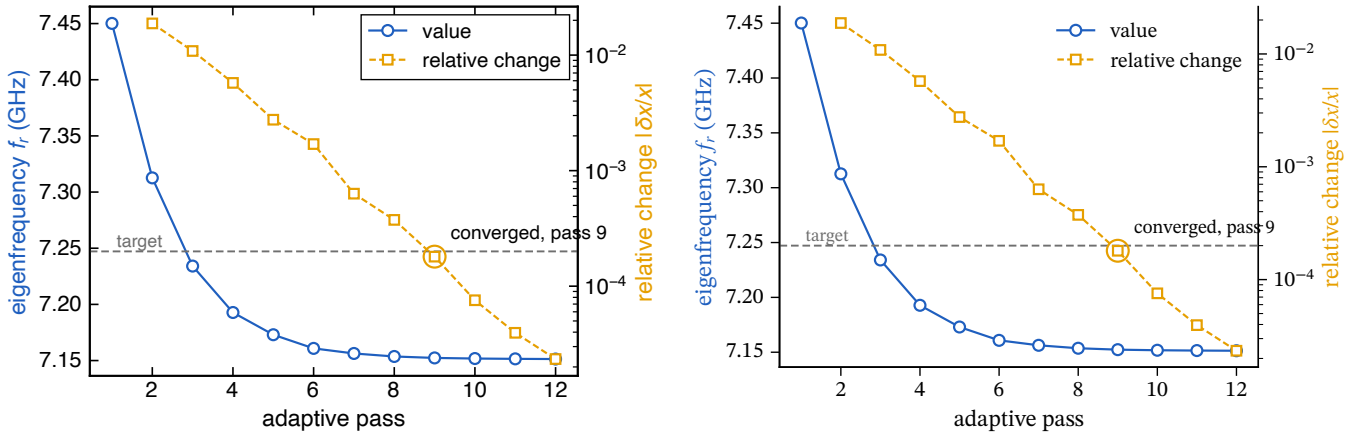
Purcell limit on T1 against qubit frequency for a 7.15 GHz readout resonator (kappa = 1.5 MHz, g = 85 MHz), without and with a Q = 25 bandpass filter (computed).



16 Convergence

`16_convergence.py`

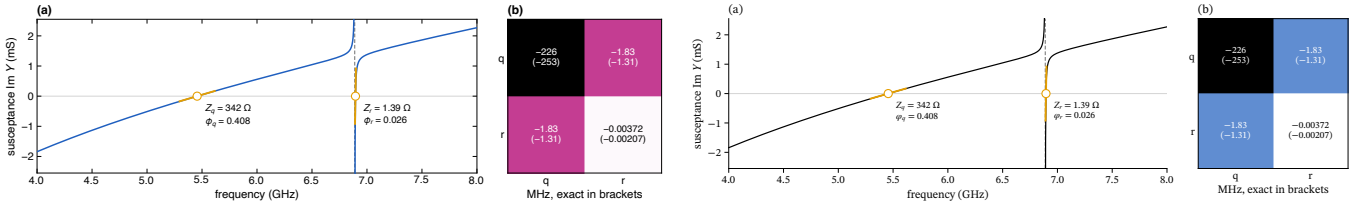
Eigenmode frequency of a readout resonator against adaptive mesh pass, with the relative change per pass and a 0.02 % target (synthetic data).



17 Black-box quantization

`17_black_box.py`

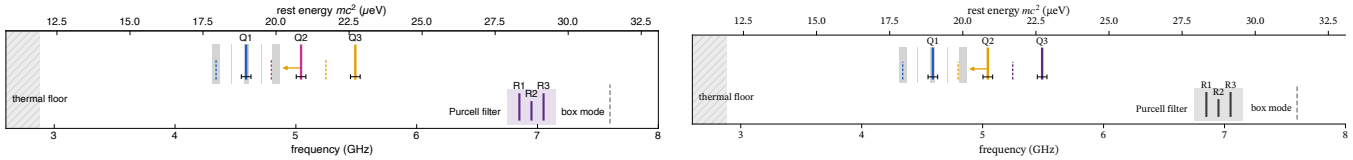
Black-box quantization of the transmon and readout resonator of example 72: (a) the susceptance the junction sees, whose rising zeros are the modes and whose slopes set their impedances; (b) the first-order Kerr matrix, with exact diagonalization of the same circuit in brackets (computed).



18 Frequency lines

`18_frequency_lines.py`

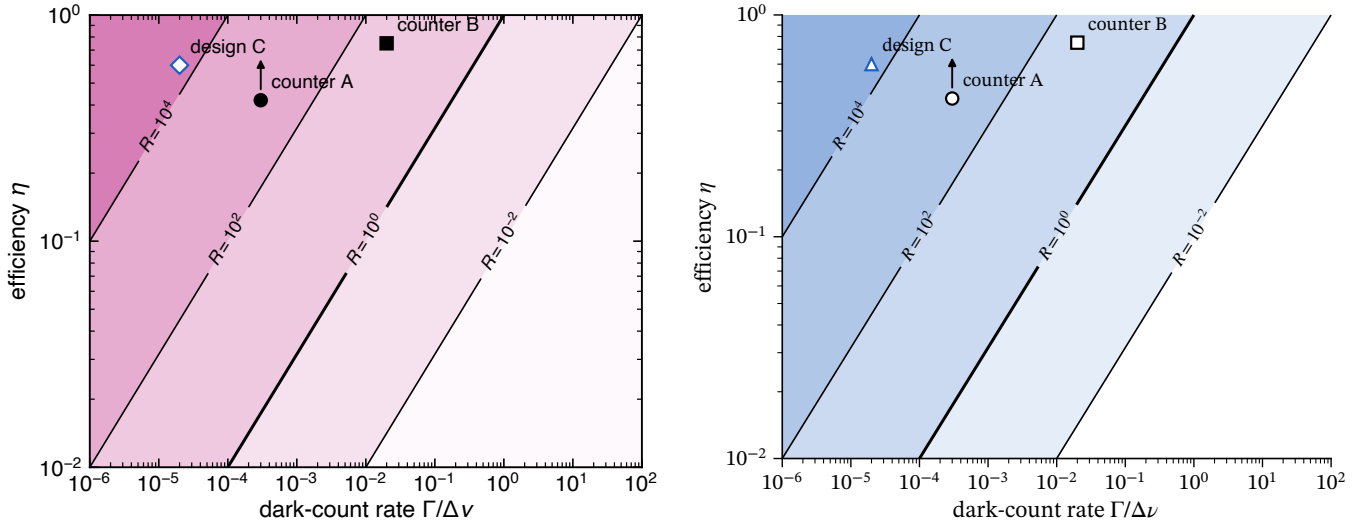
Frequency plan of a three-qubit sensor chip on one axis: f01 and f12 lines with their fabrication scatter, one flux scan, the collision windows of the tunable qubit, the readout resonators in a Purcell passband, the 20 mK thermal floor and the rest-energy axis (designed values).



19 Design map

`19_design_map.py`

Scan-rate gain R = eta^2 dnu/Gamma of a photon counter over a quantum-limited amplifier on the plane of dark-count rate (over the signal linewidth) and efficiency, with R = 1 drawn heavier and three illustrative devices, one an efficiency lower bound (computed map, synthetic points).

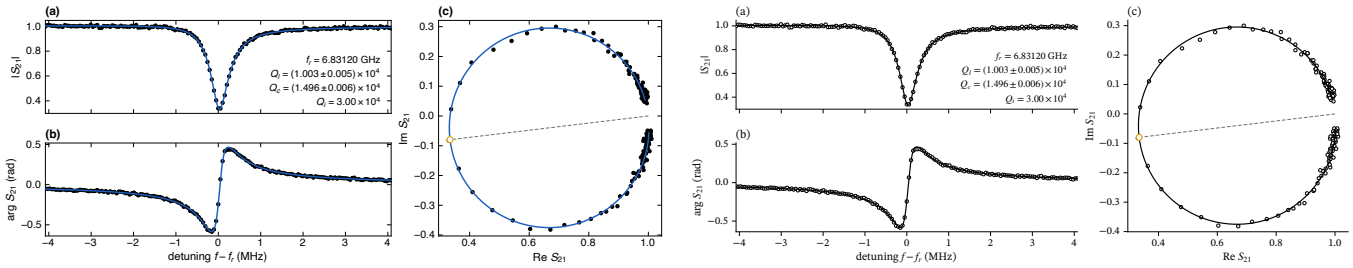


Readout

20 S21 of a resonator

`20_s21_resonator.py`

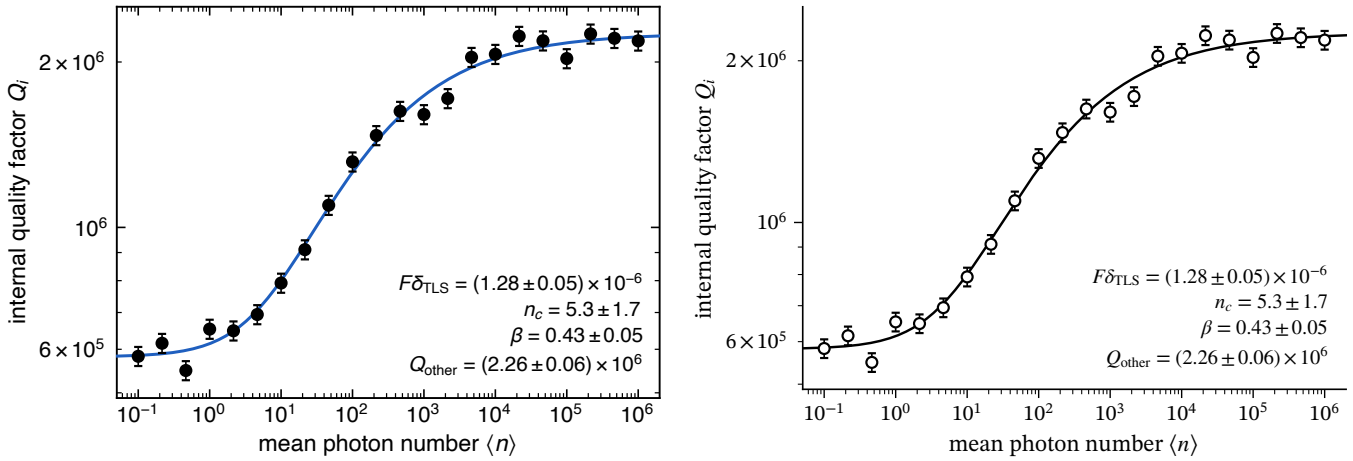
Notch-resonator transmission as magnitude, phase and circle, with one complex fit and the cable delay, amplitude and phase calibrated out (synthetic data).



21 Quality factor vs power

`21_quality_vs_power.py`

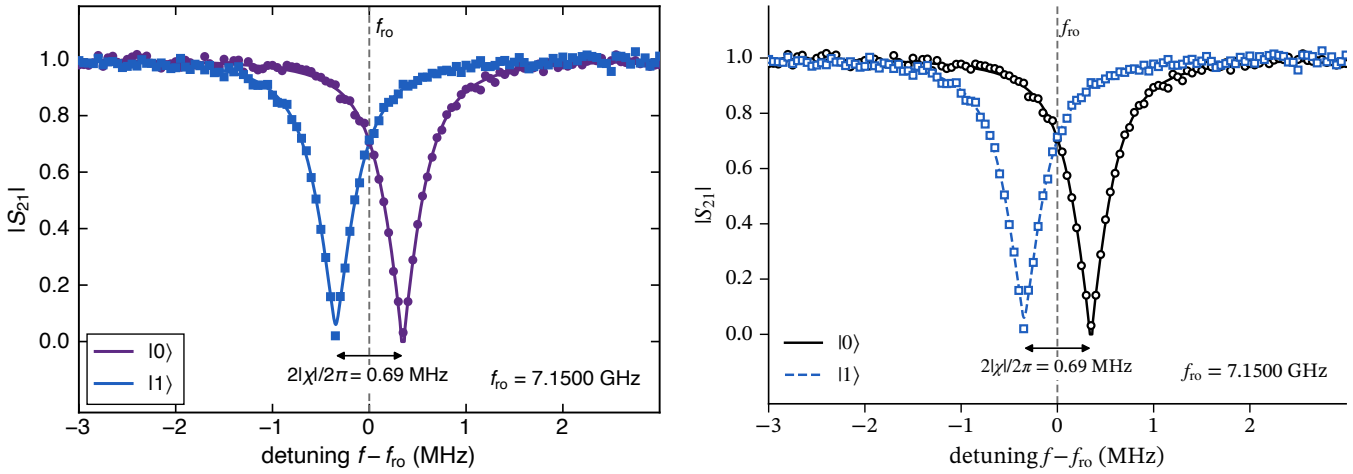
Internal quality factor of a resonator against mean photon number, with a fit of the saturable two-level-system model (synthetic data).



22 Dispersive readout

`22_dispersive_readout.py`

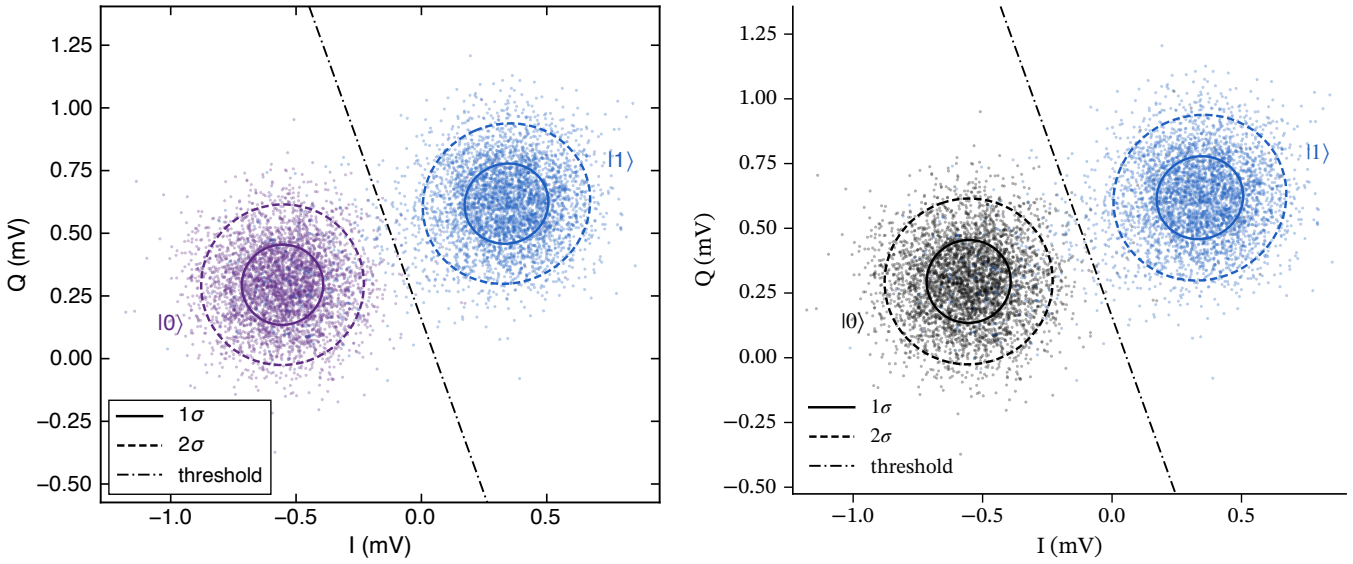
Dispersive readout: resonator transmission with the qubit in |0> and in |1>, the 2chi shift computed for g = 85 MHz and Delta = -2.09 GHz, and the readout frequency between the two dips (synthetic data).



23 IQ blobs

`23_iq_blobs.py`

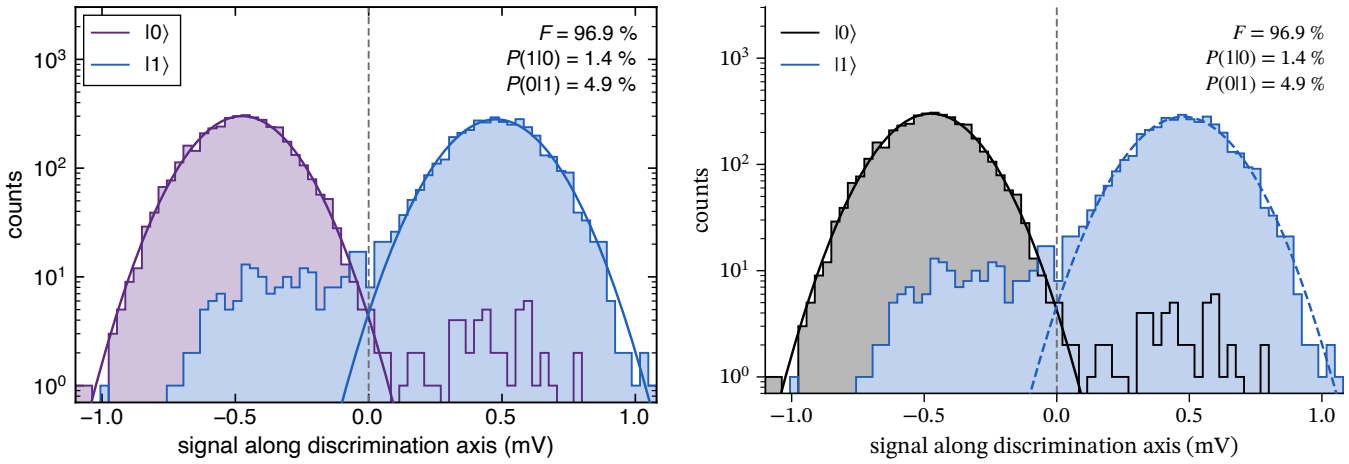
Single-shot readout in the IQ plane: 4000 shots per prepared state with 1 and 2 sigma contours and the discrimination line; the |1> shots include T1 decay during the 1.5 us integration (synthetic data).



24 Readout histogram

`24_readout_histogram.py`

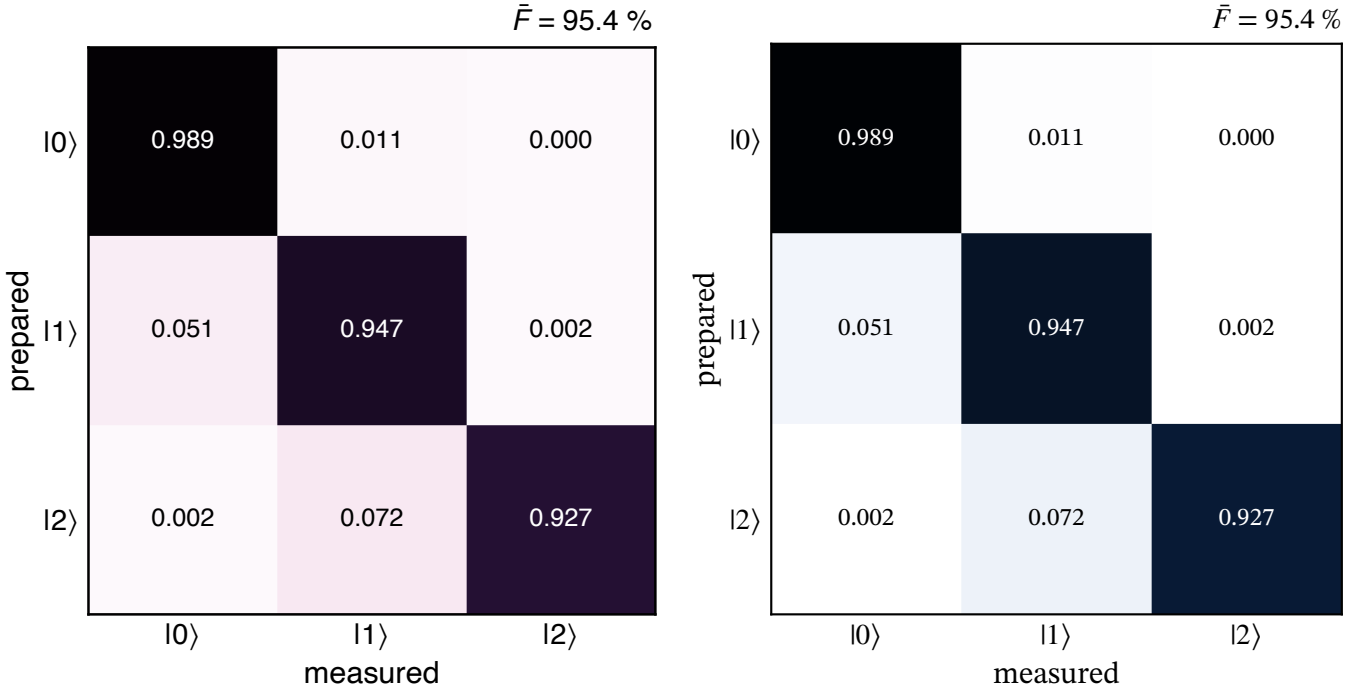
Single-shot histograms projected on the discrimination axis, with Gaussian fits to each peak and the assignment fidelity; the tail between the peaks is T1 decay during readout (synthetic data).



25 Assignment matrix

`25_assignment_matrix.py`

Three-state assignment matrix from 5000 simulated shots per prepared state, assigned to the nearest IQ centre; decay during readout fills the lower triangle (synthetic data).

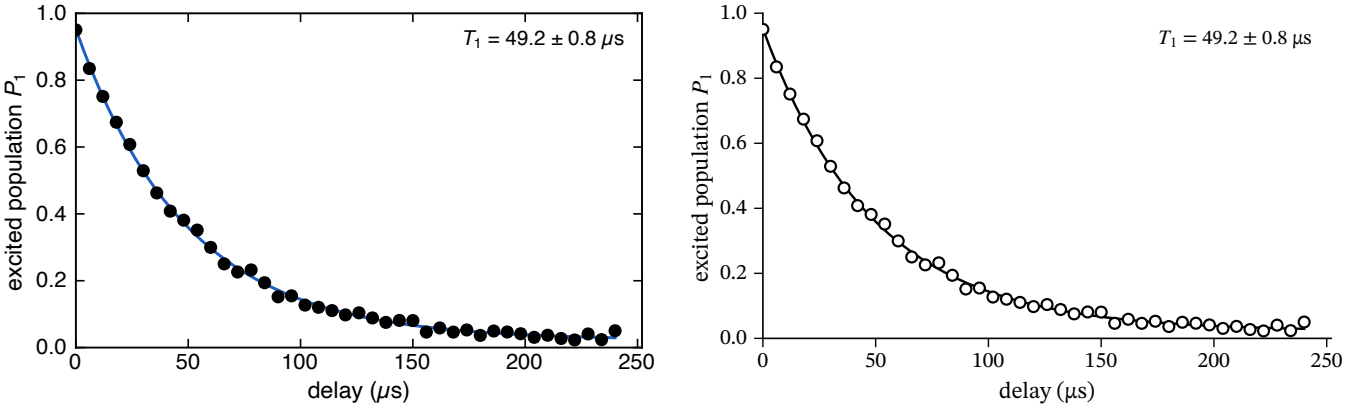


Time domain

30 T1 decay

`30_t1_decay.py`

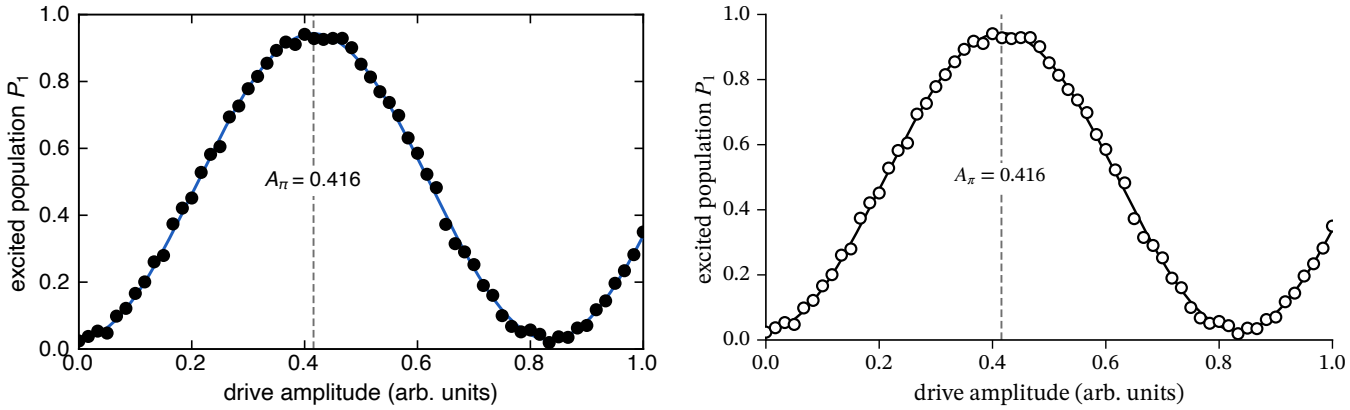
Energy relaxation: excited-state population against delay with an exponential fit (synthetic data, T1 = 48 us).



31 Power Rabi

`31_rabi.py`

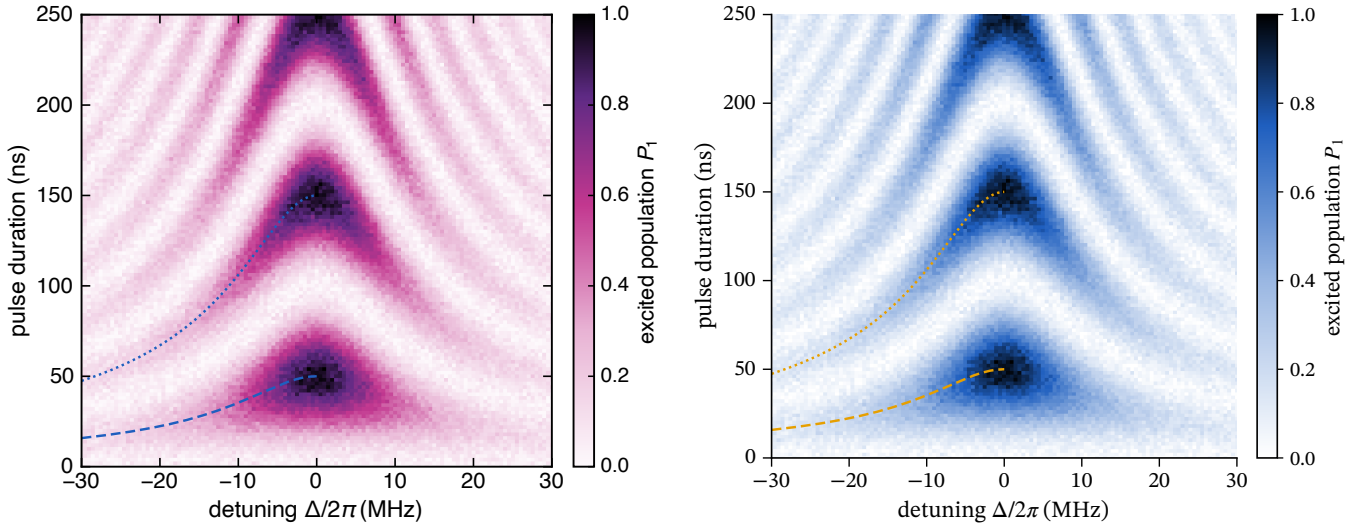
Power Rabi: excited population against drive amplitude for a 40 ns pulse, with a cosine fit and the pi-pulse amplitude (synthetic data).



32 Rabi chevron

`32_chevron.py`

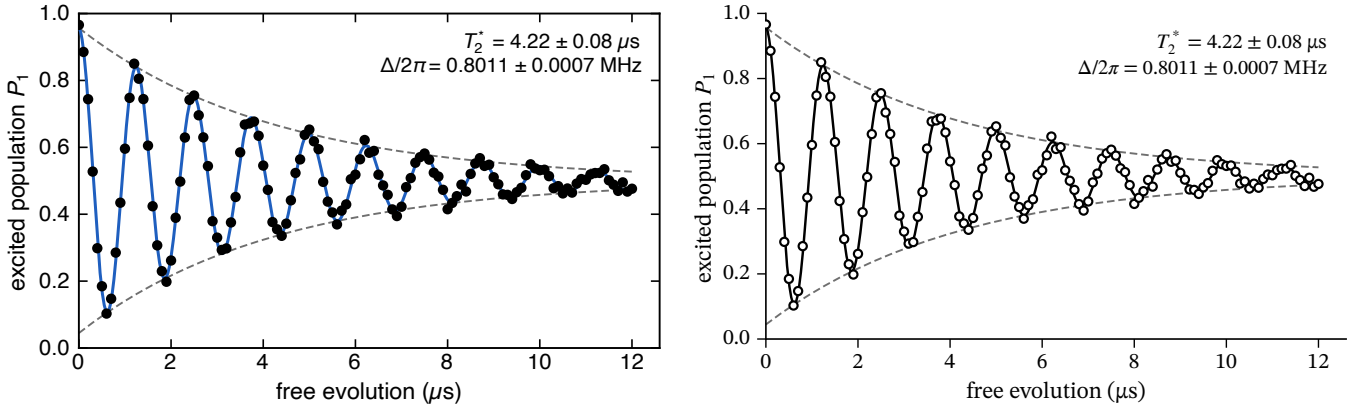
Rabi chevron: excited population against drive detuning and pulse duration at a 10 MHz Rabi rate, with the first two population maxima of the model drawn over the left half (synthetic data).



33 Ramsey fringes

`33_ramsey.py`

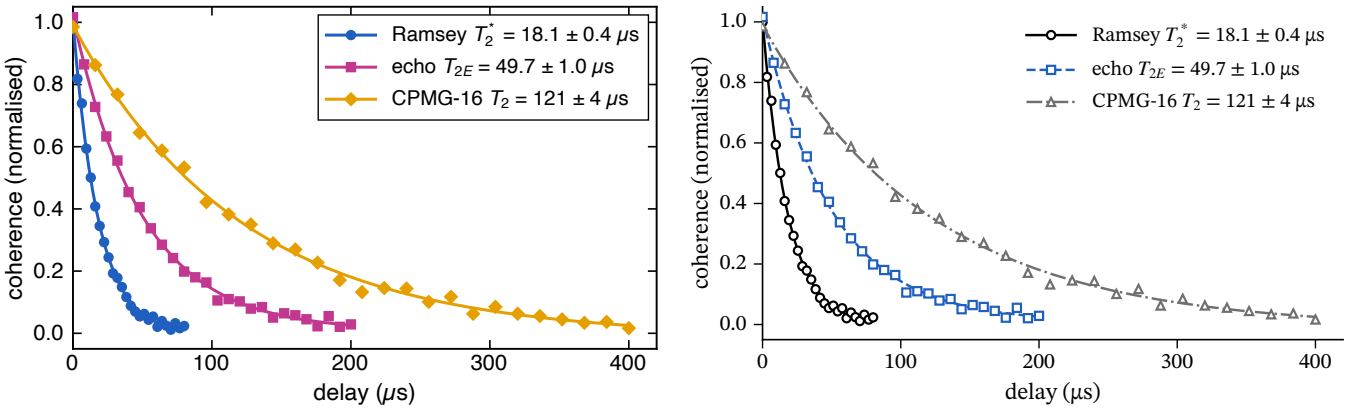
Ramsey fringes at 0.8 MHz detuning with a damped-cosine fit and its envelope (synthetic data, T2* = 4.2 us).



34 Coherence decays

`34_coherence_decays.py`

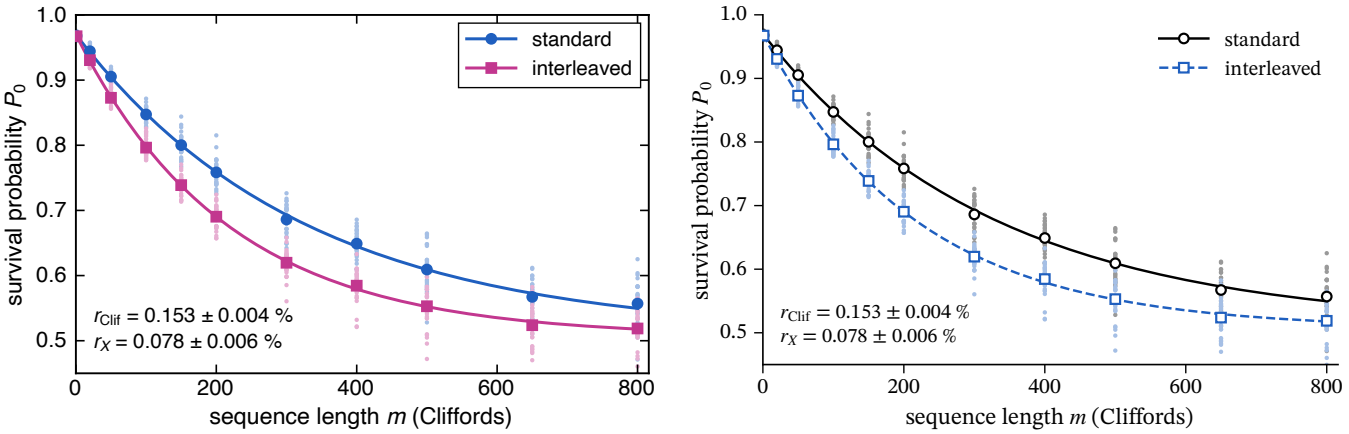
Ramsey envelope, Hahn echo and CPMG-16 decays of one qubit on one axis, each with its exponential fit and time constant (synthetic data).



35 Randomized benchmarking

`35_rb.py`

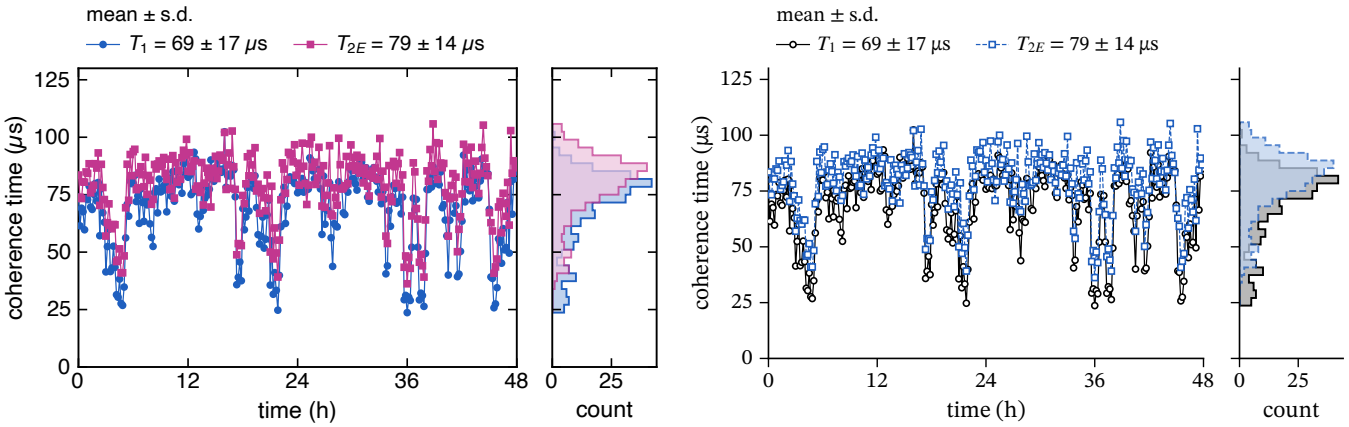
Standard and interleaved randomized benchmarking of an X gate, 30 random sequences per length, with the error per Clifford and the interleaved gate error (synthetic data).



36 Coherence over time

`36_coherence_timeseries.py`

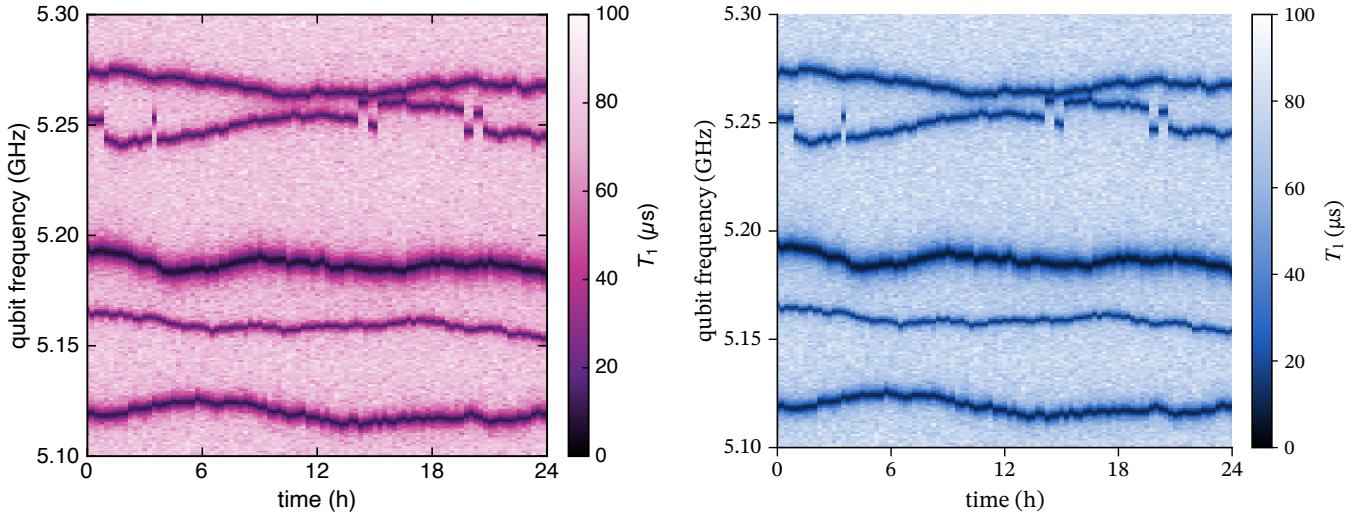
T1 and echo T2 measured every 10 minutes for 48 hours, with their histograms; T1 drops whenever a wandering defect drifts onto the qubit frequency (synthetic data).



37 TLS spectral diffusion

`37_tls_map.py`

T1 against qubit frequency over 24 hours; five defects show as Lorentzian T1 dips whose frequencies wander, and one switches between two frequencies (synthetic data).

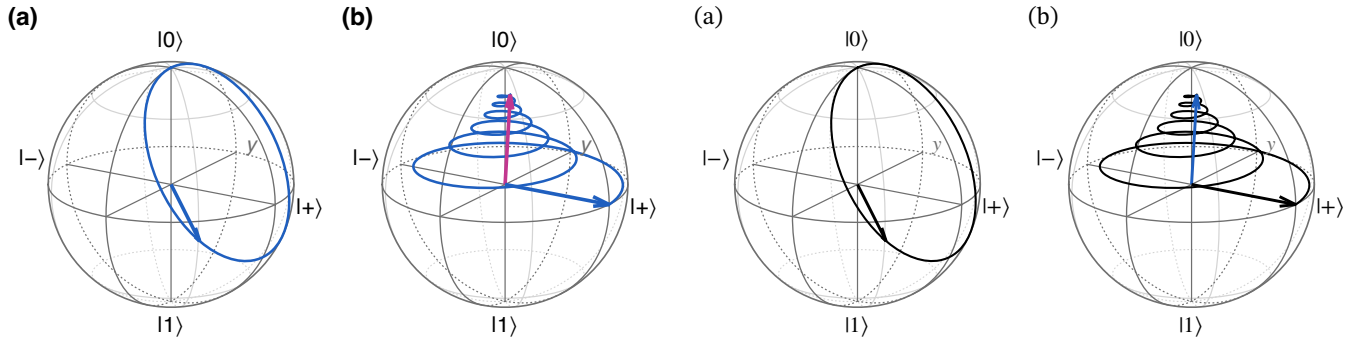


States

40 Bloch sphere

`40_bloch_sphere.py`

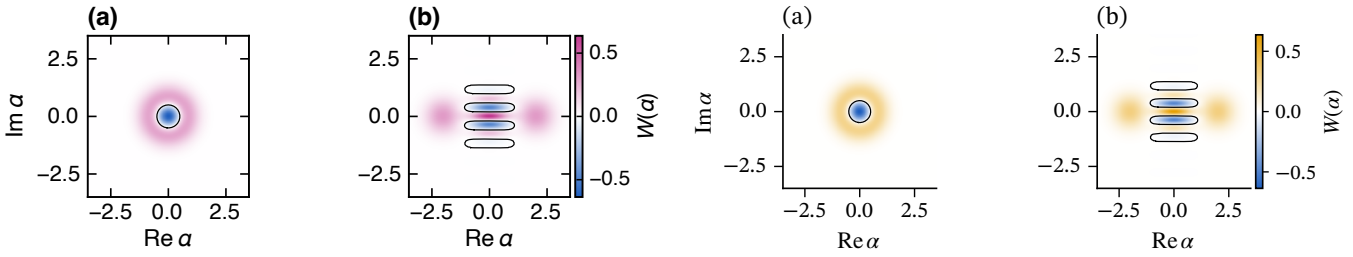
Bloch sphere: (a) a detuned Rabi drive (Delta = 0.6 Omega) tilts the precession cone off the pole; (b) free precession from |+> spirals to |0> as T2 and T1 shrink the vector (computed).



41 Wigner functions

`41_wigner.py`

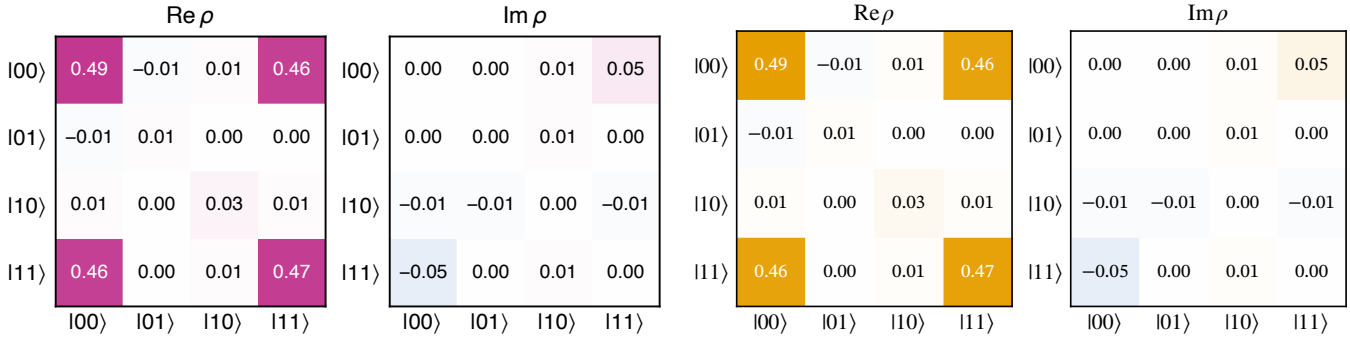
Wigner functions of (a) the Fock state |1> and (b) an even cat state with alpha = 2, on one colour scale of +-2/pi (computed).



42 Density matrix

42_density_matrix.py

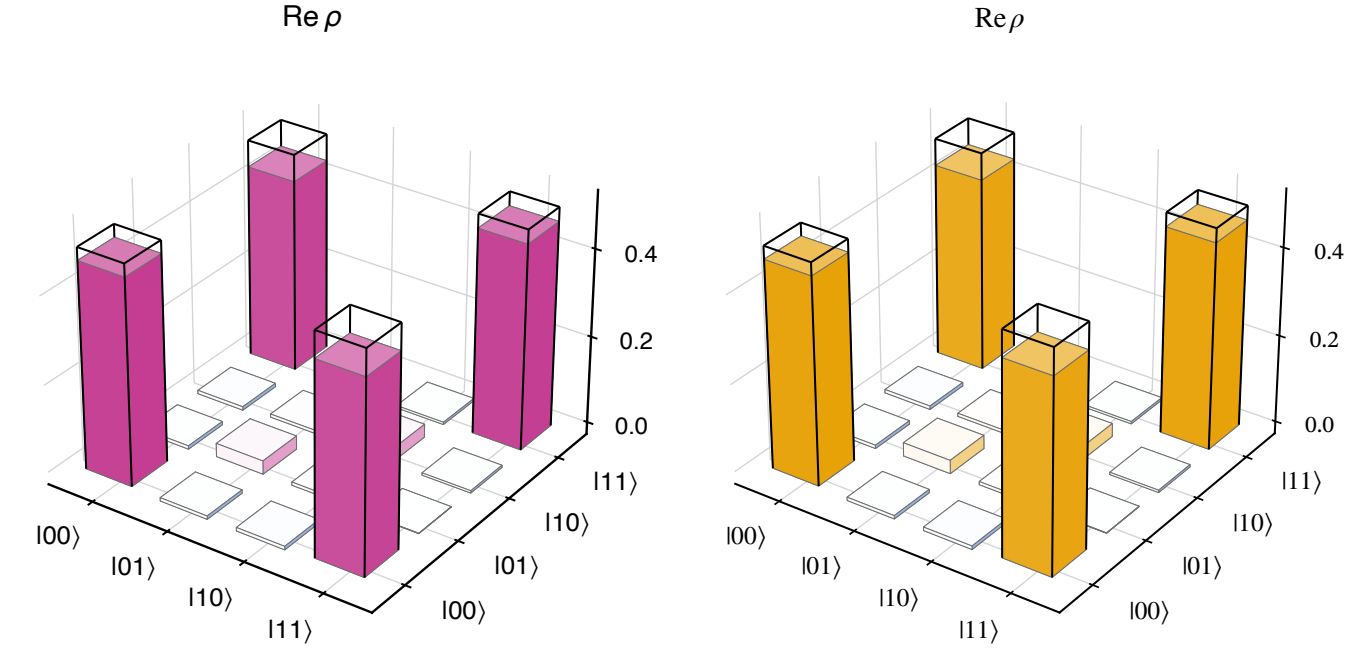
Two-qubit tomography of the Bell state (|00> + |11>)/sqrt(2): real and imaginary parts of the reconstructed density matrix (synthetic data).



43 Density matrix in 3D

43_density_bars3d.py

Real part of the reconstructed Bell-state density matrix as 3D bars, with the ideal state as wireframe outlines (synthetic data).

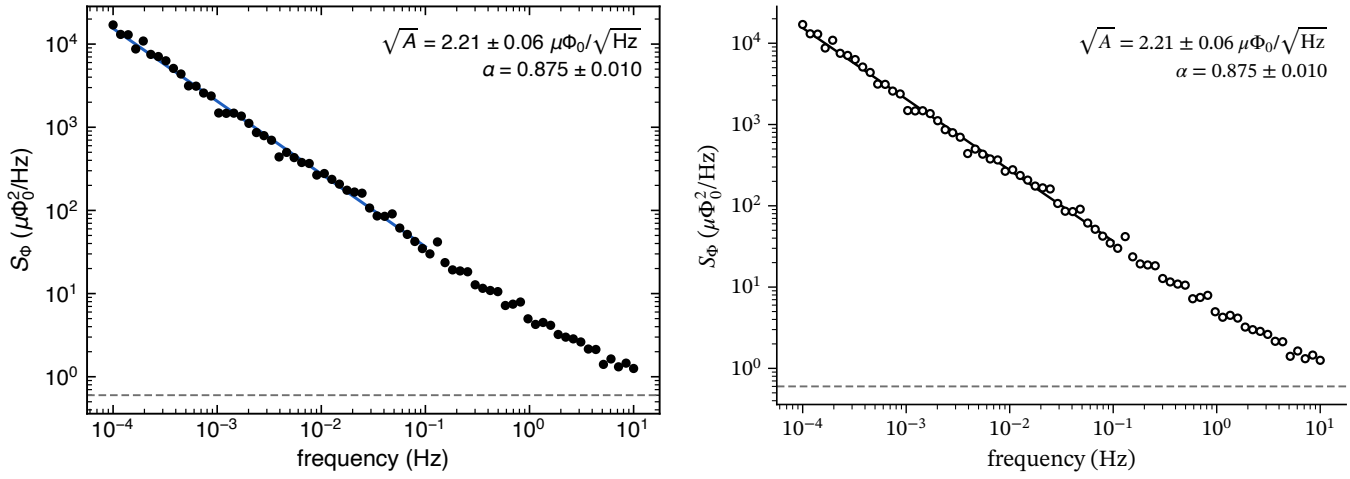


Noise and errors

50 Noise spectral density

50_noise_psd.py

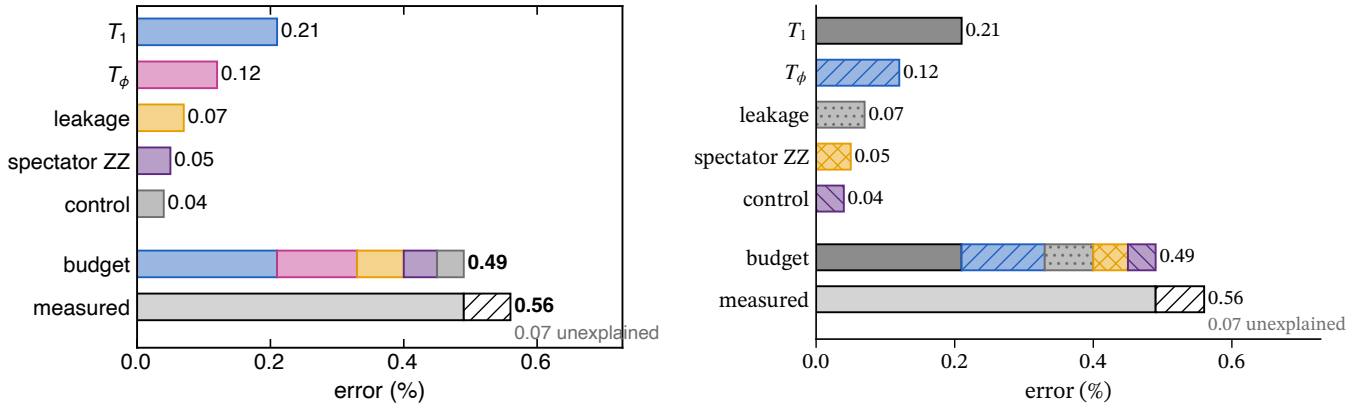
Flux-noise spectral density with a power-law fit, amplitude at 1 Hz and a white floor (synthetic data).



51 Error budget

`51_error_budget.py`

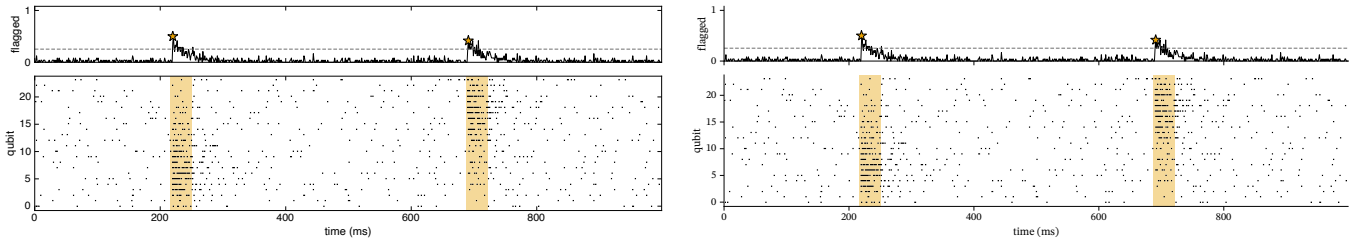
CZ-gate error budget: contributions, their sum, and the measured error with the unexplained part hatched.



52 Correlated error bursts

`52_burst_raster.py`

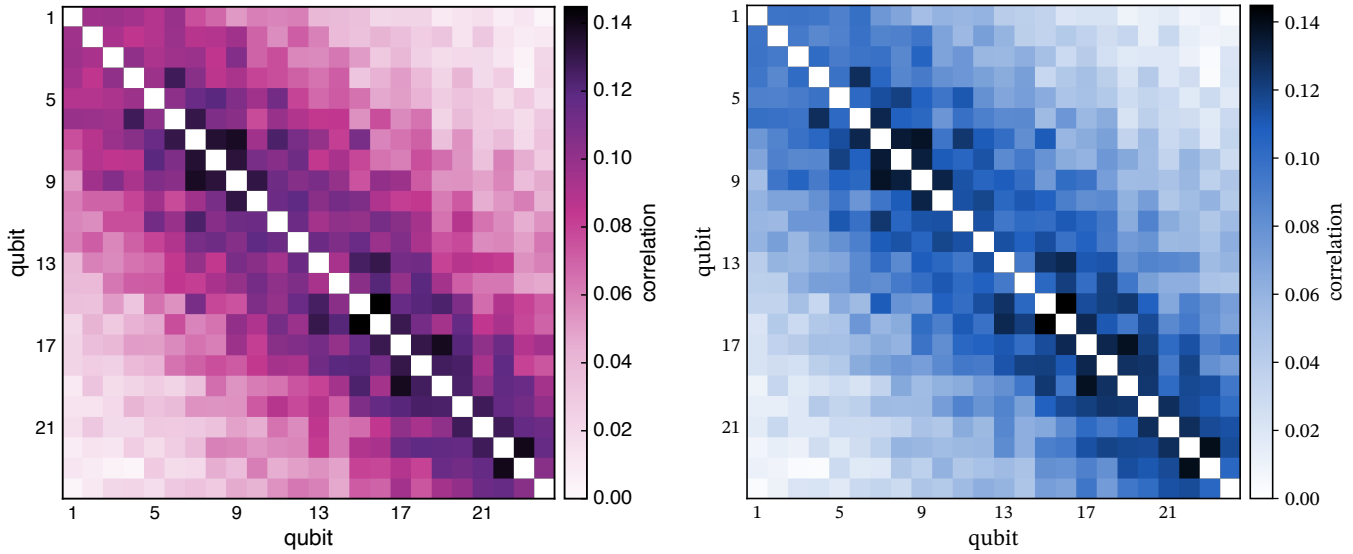
Correlated error bursts: detections per qubit over time, with the flagged fraction and burst marks above (synthetic data from a toy burst model).



53 Error correlations

`53_error_correlation.py`

Pairwise correlation of error detections across a 24-qubit chain; bursts correlate neighbours (synthetic data from a toy burst model).

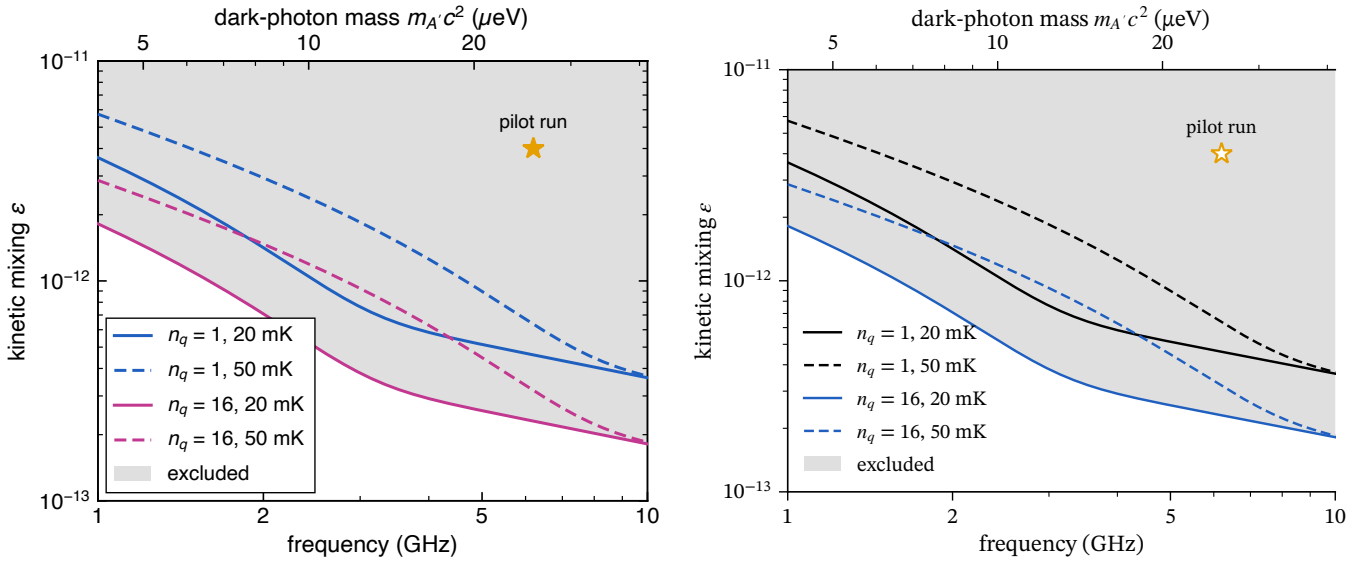


Dark matter

59 Dark-photon reach

`59_reach.py`

Kinetic-mixing reach of a qubit dark-photon search against frequency for one and sixteen sensors at 20 and 50 mK, from an illustrative counting model with thermal dark counts, an excluded region and a synthetic measured limit, with the dark-photon mass on the top axis (computed).

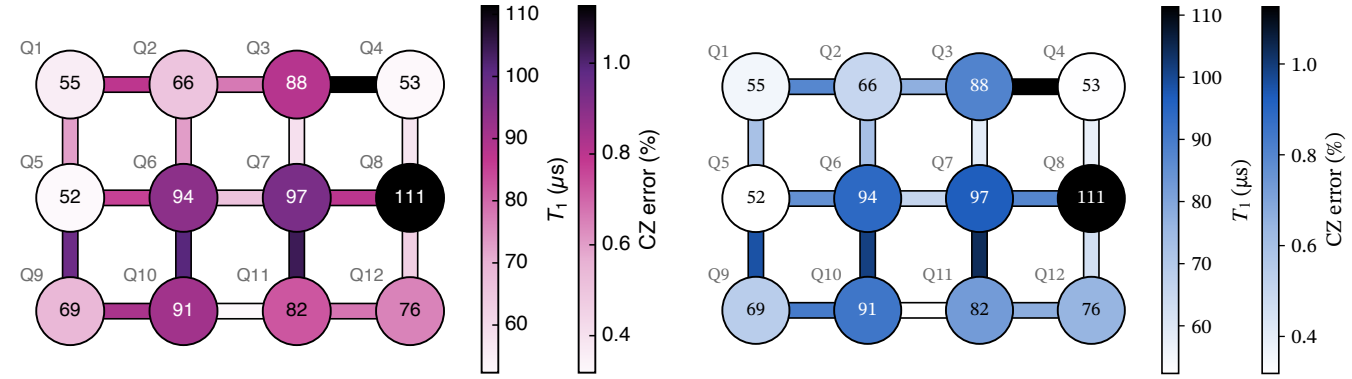


Processor

60 Device map

`60_device_map.py`

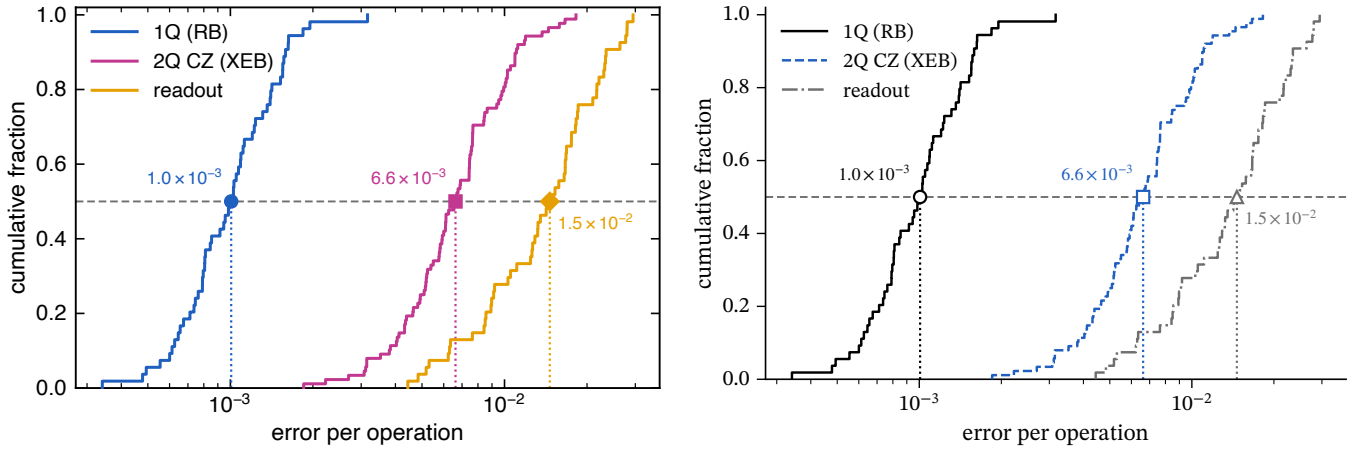
Device map of a 3 x 4 qubit grid with T1 on the qubits and CZ error on the couplers (synthetic data).



61 Error distributions

`61_error_cdf.py`

Cumulative distributions of single-qubit, CZ and readout errors across a 54-qubit processor, medians marked (synthetic data).

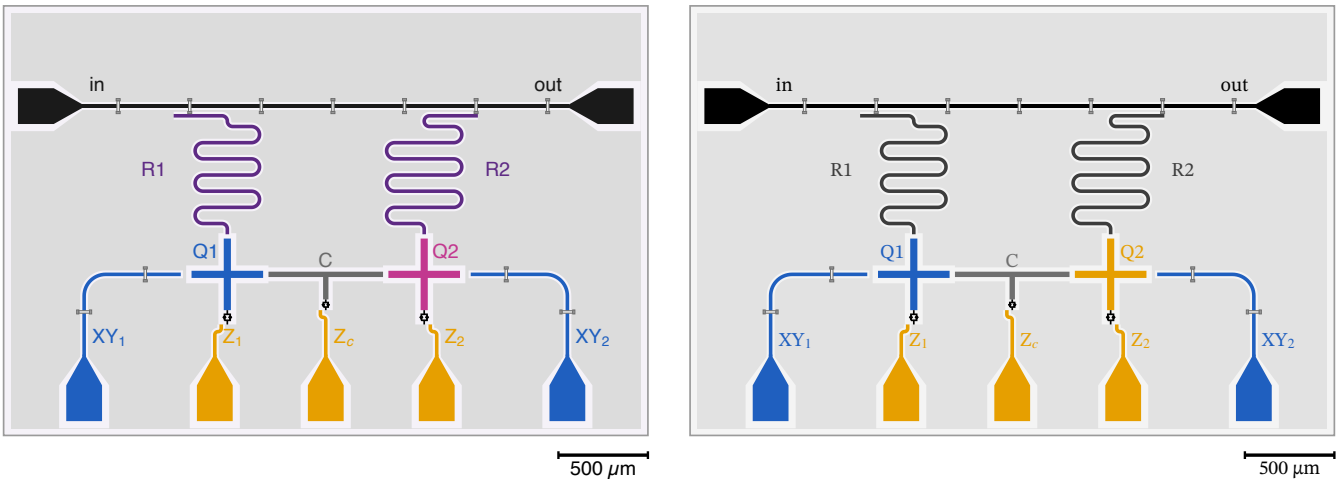


Schematics

70 Chip plan

`70_chip_plan.py`

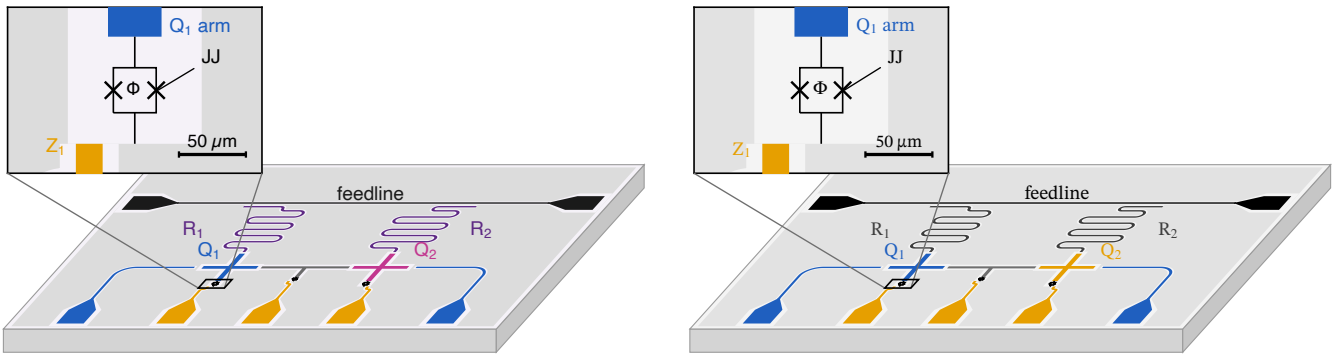
Plan view of a two-qubit chip: Xmons joined by a tunable coupler, readout resonators on a feedline, drive and flux lines, airbridges.



71 Chip on a slab

`71_chip_oblique.py`

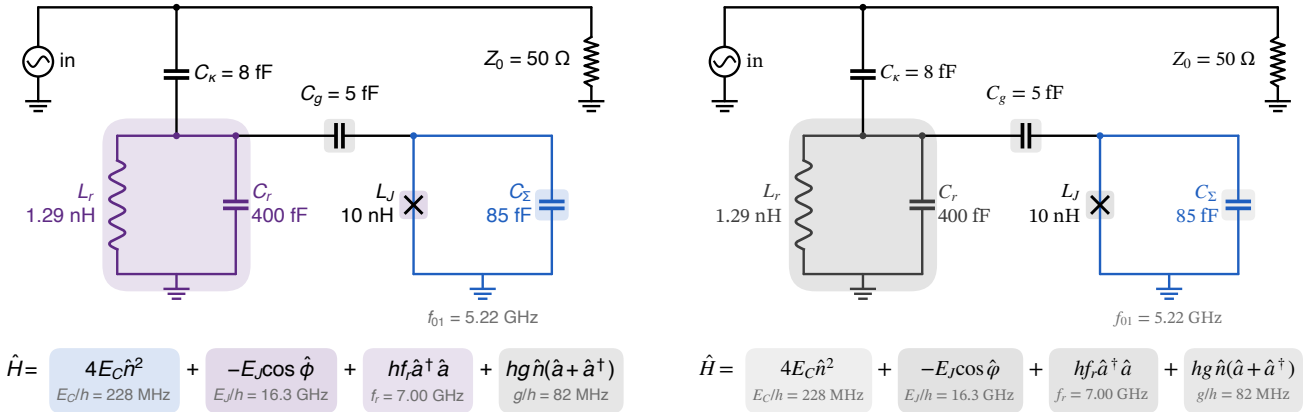
The two-qubit chip of example 70 laid on a slab by one rigid oblique map, with a plan-view zoom on the SQUID of Q1.



72 Transmon and readout circuit

`72_lumped_circuit.py`

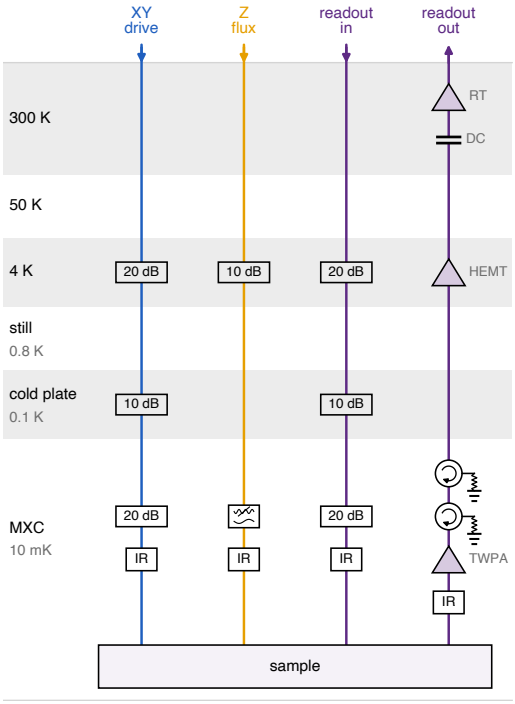
Lumped circuit of a transmon coupled to a quarter-wave readout resonator on a feedline, each Hamiltonian term on the tile of the element that produces it.

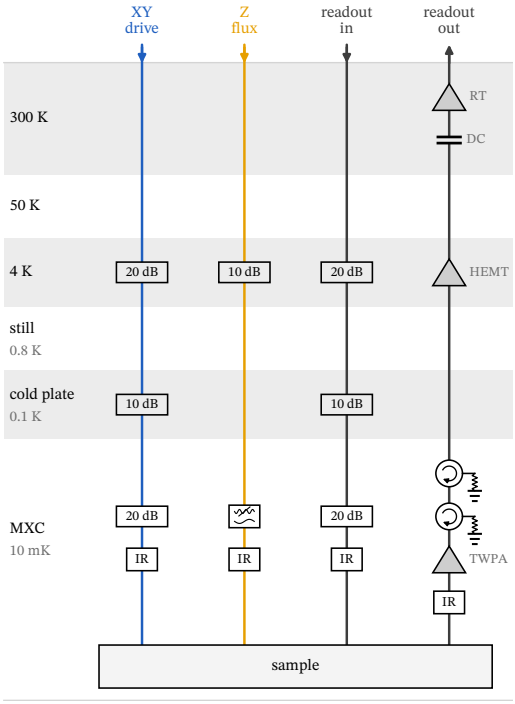


73 Cryostat wiring

`73_cryostat_wiring.py`

Wiring of a dilution refrigerator: attenuated and filtered XY, Z and readout-input lines, and a readout output through isolators, a TWPA and a HEMT.

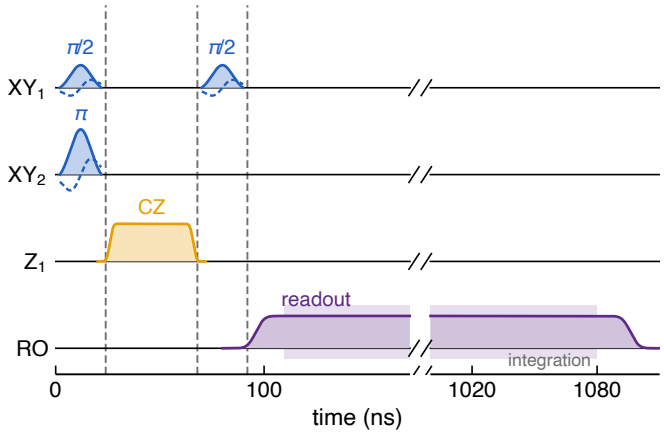


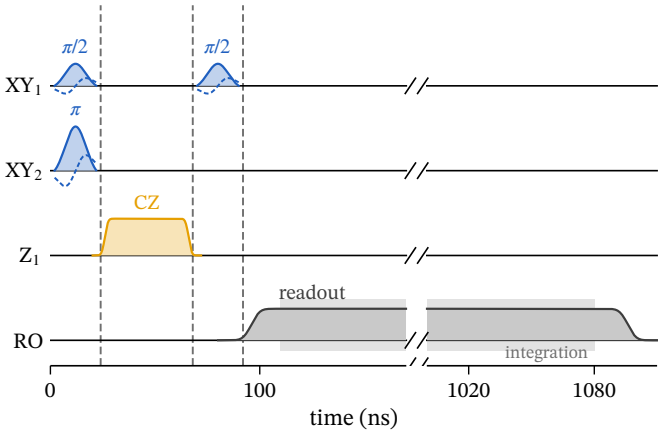


74 Pulse sequence

`74_pulse_sequence.py`

Pulse sequence of a conditional-phase measurement: DRAG pulses on two qubits, a CZ flux pulse and a dispersive readout with its integration window.

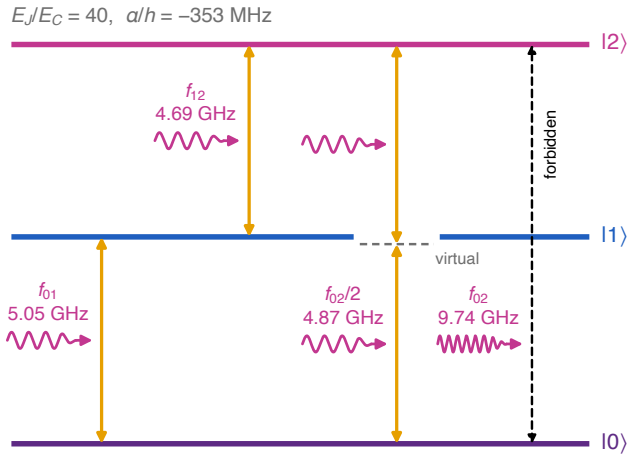


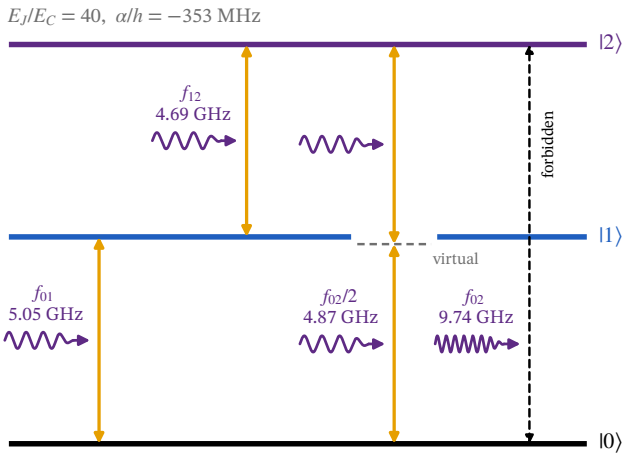


75 Level ladder and transitions

`75_level_ladder.py`

Transmon ladder (E_J/E_C = 40) to scale, with drive photons for f01, f12 and the two-photon f02/2, and the parity-forbidden single-photon 0-2 transition.

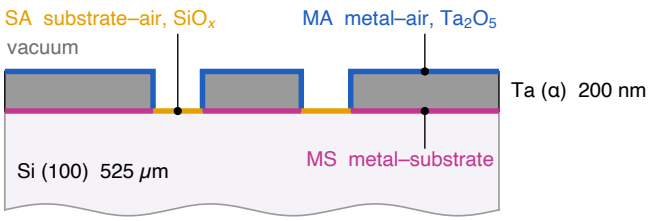


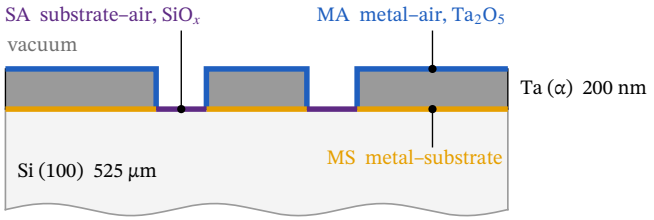


76 Film stack

`76_film_stack.py`

Cross-section of a tantalum CPW on silicon with its metal-air, metal-substrate and substrate-air interfaces marked.

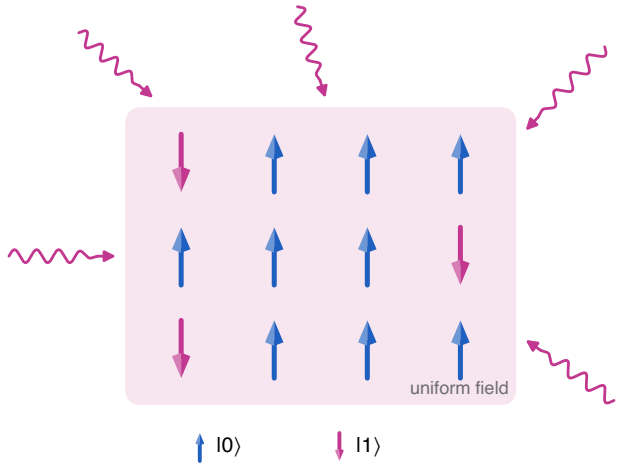


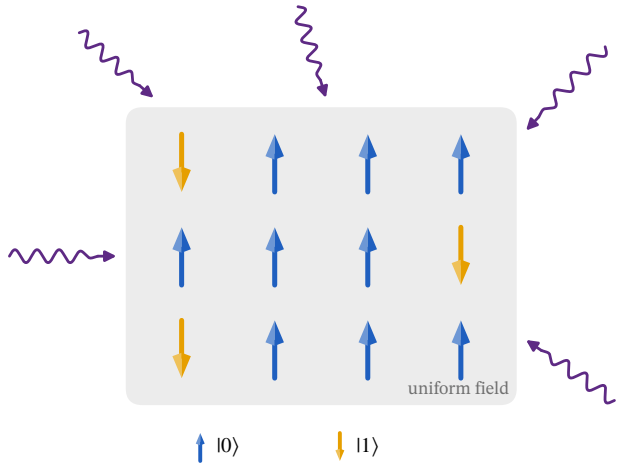


77 Qubit array in a uniform field

`77_spin_array.py`

A 3 x 4 qubit array drawn as spins in a uniform field (the wash), with photons arriving from several directions and three qubits excited.

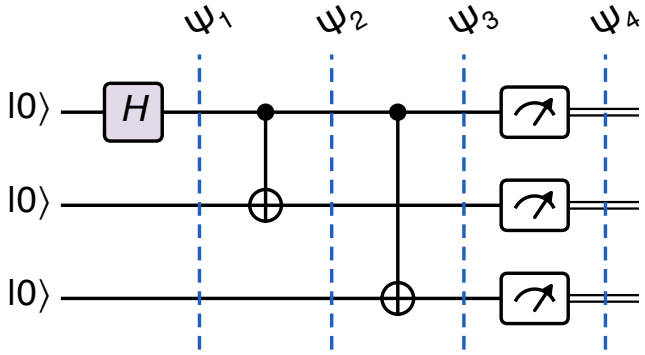


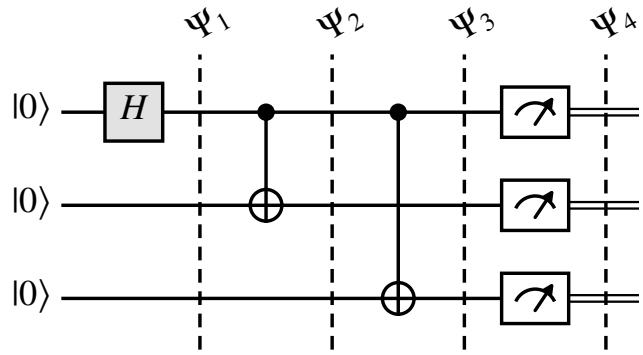


78 Quantum circuit

`78_quantum_circuit.py`

GHZ-state preparation: a Hadamard and two CNOTs entangle three qubits, then all three are measured; dashed slices mark the states Psi_1 to Psi_4.

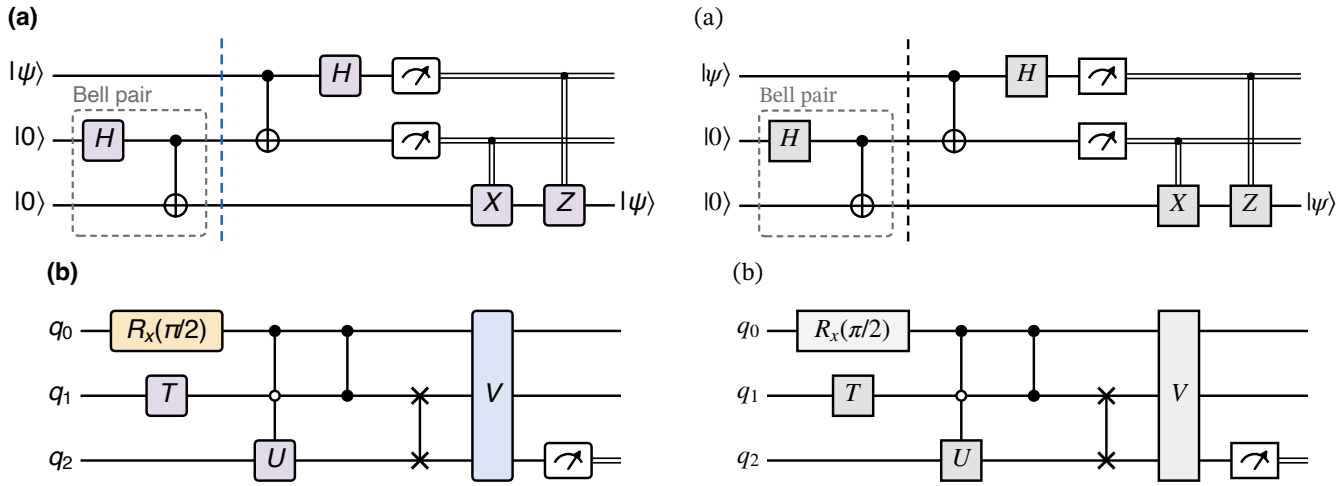




79 Teleportation and gate glyphs

`79_teleportation.py`

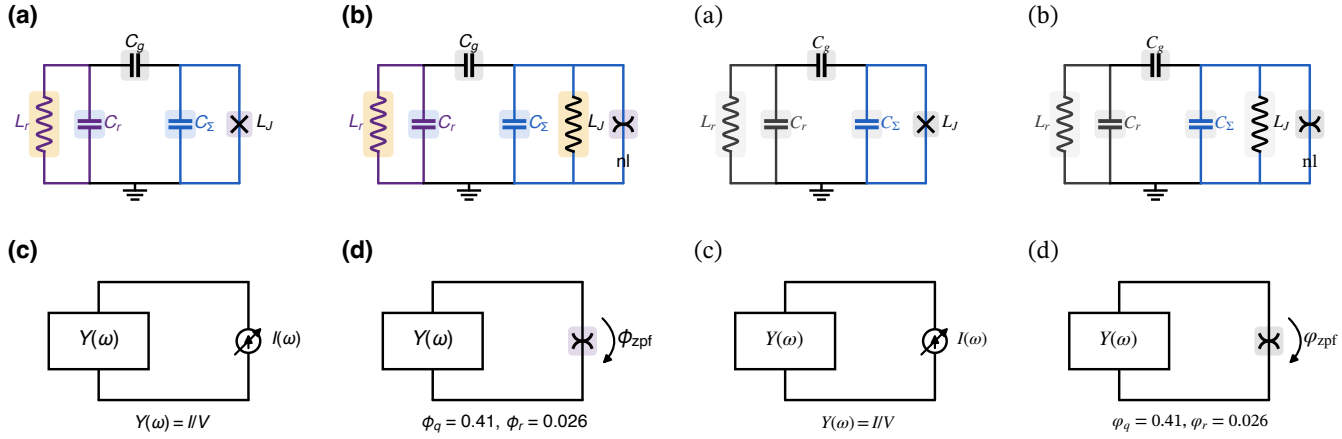
(a) Teleportation: a Bell pair, a Bell measurement and classically controlled X and Z corrections. (b) The other gate glyphs: rotation, controlled-U with an open control, CZ, SWAP and a three-qubit box.



80 Black-box circuit

`80_black_box_circuit.py`

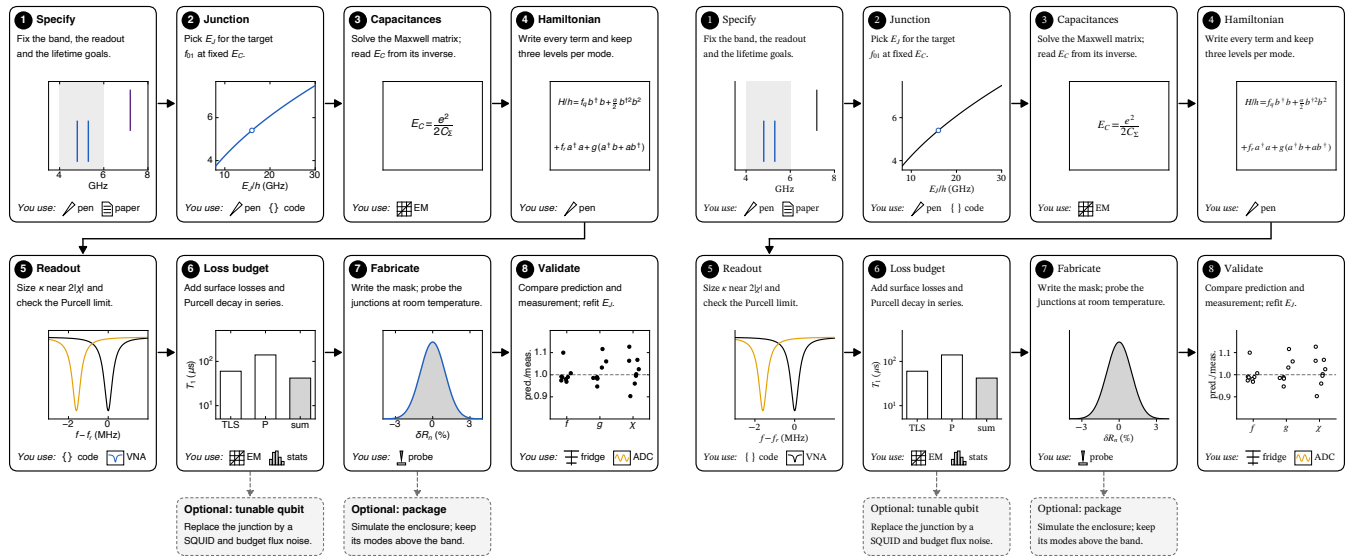
Black-box quantization in four steps for the device of example 72: (a) the transmon coupled to its readout resonator; (b) the junction split into its linear inductance and its nonlinear remainder; (c) the linear part as one admittance, probed by a swept current source; (d) that admittance beside the nonlinear element, whose zero-point phase each mode sets (computed).



81 Recipe grid

`81_recipe_grid.py`

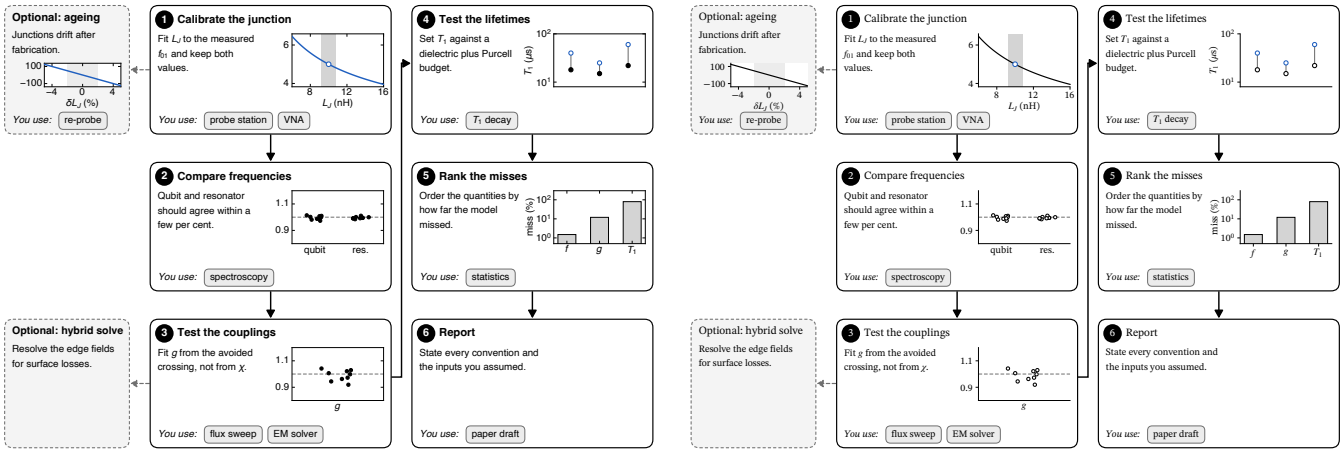
A generic transmon design recipe in eight numbered steps: each box holds instruction lines, a mini-plot from toy models and the tools the step uses, arrows run through the grid and two optional steps sit in dashed boxes below.



82 Step guide

82_step_guide.py

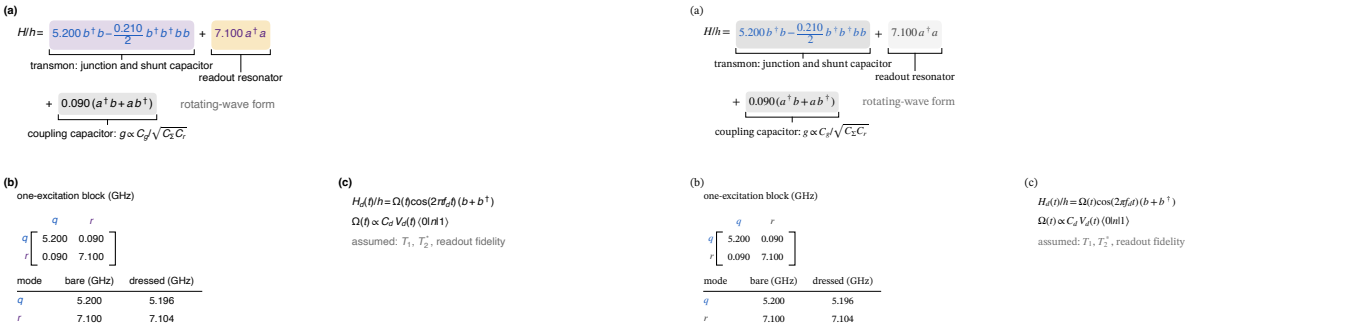
A generic validation guide in two columns: the numbered steps run down each column with a mini-plot beside the text and the tools as labelled chips, and two optional steps sit in dashed side boxes joined to their step.



83 Hamiltonian terms

83_hamiltonian_terms.py

A generic transmon-resonator Hamiltonian: (a) every term on the tile of the circuit element that produces it, bracketed by origin, (b) the one-excitation block as a bracketed matrix with its bare and dressed frequencies, (c) the drive term and the inputs that stay assumed (illustrative numbers).

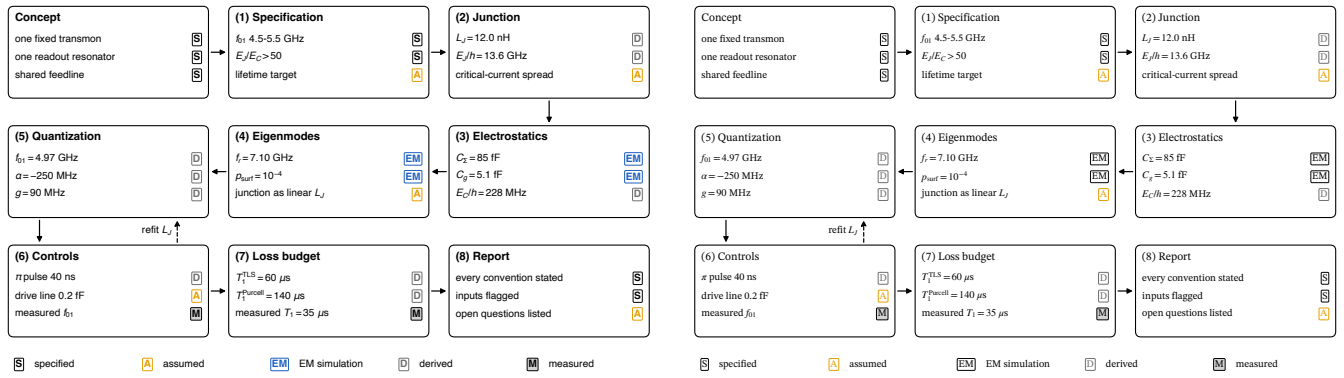


(a)
$H/h = 5.200\, b^\dagger b - \frac{0.210}{2} b^\dagger b^\dagger b b + 7.100\, a^\dagger a$
transmon: junction and shunt capacitor
readout resonator
$+ 0.090\,(a^\dagger b + a b^\dagger)$
rotating-wave form
coupling capacitor: $g \propto C_g/\sqrt{C_\Sigma C_r}$
(b)
one-excitation block (GHz)
q r
q 5.200 0.090
r 0.090 7.100
mode bare (GHz) dressed (GHz)
q 5.200 5.196
r 7.100 7.104
(c)
$H_d(t)/h = \Omega(t)\cos(2\pi f_d t)\,(b + b^\dagger)$
$\Omega(t) \propto C_d\, V_d(t)\,\langle 0|n|1\rangle$
assumed: T_1, T_2^*, readout fidelity

84 Design flow

84_design_flow.py

A generic chip design flow on a three-by-three grid: each box lists its quantities with a status tag (specified, assumed, simulated, derived, measured), arrows snake through the steps and a dashed arrow carries the measured frequency back to the junction (illustrative numbers).

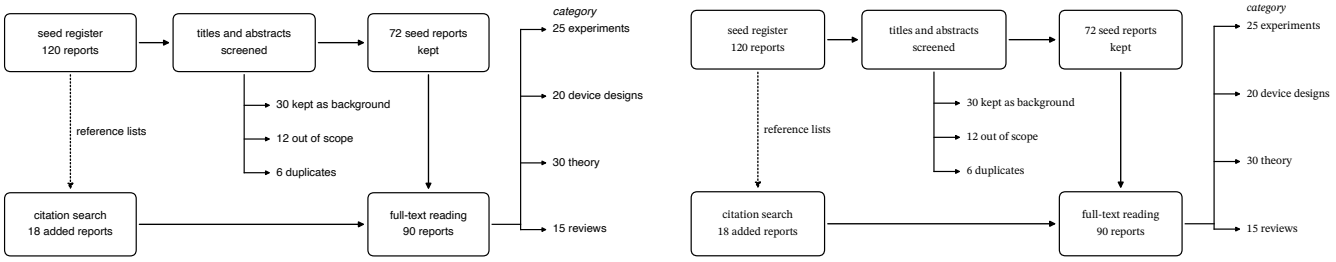


Concept
one fixed transmon S
one readout resonator S
shared feedline S
(1) Specification
f_{01} 4.5-5.5 GHz S
$E_J/E_C > 50$ S
lifetime target A
(2) Junction
$L_J = 12.0$ nH D
$E_J/h = 13.6$ GHz D
critical-current spread A
(3) Electrostatics
$C_\Sigma = 85$ fF EM
$C_g = 5.1$ fF EM
$E_C/h = 228$ MHz D
(4) Eigenmodes
$f_r = 7.10$ GHz EM
$p_{surf} = 10^{-4}$ EM
junction as linear L_J A
(5) Quantization
$f_{01} = 4.97$ GHz D
$\alpha = -250$ MHz D
$g = 90$ MHz D
refit L_J
(6) Controls
π pulse 40 ns D
drive line 0.2 fF A
measured f_{01} M
(7) Loss budget
$T_1^{TLS} = 60\ \mu s$ D
$T_1^{Purcell} = 140\ \mu s$ D
measured $T_1 = 35\ \mu s$ M
(8) Report
every convention stated S
inputs flagged S
open questions listed A
S specified
A assumed
EM EM simulation
D derived
M measured

85 Screening flow

`85_screening_flow.py`

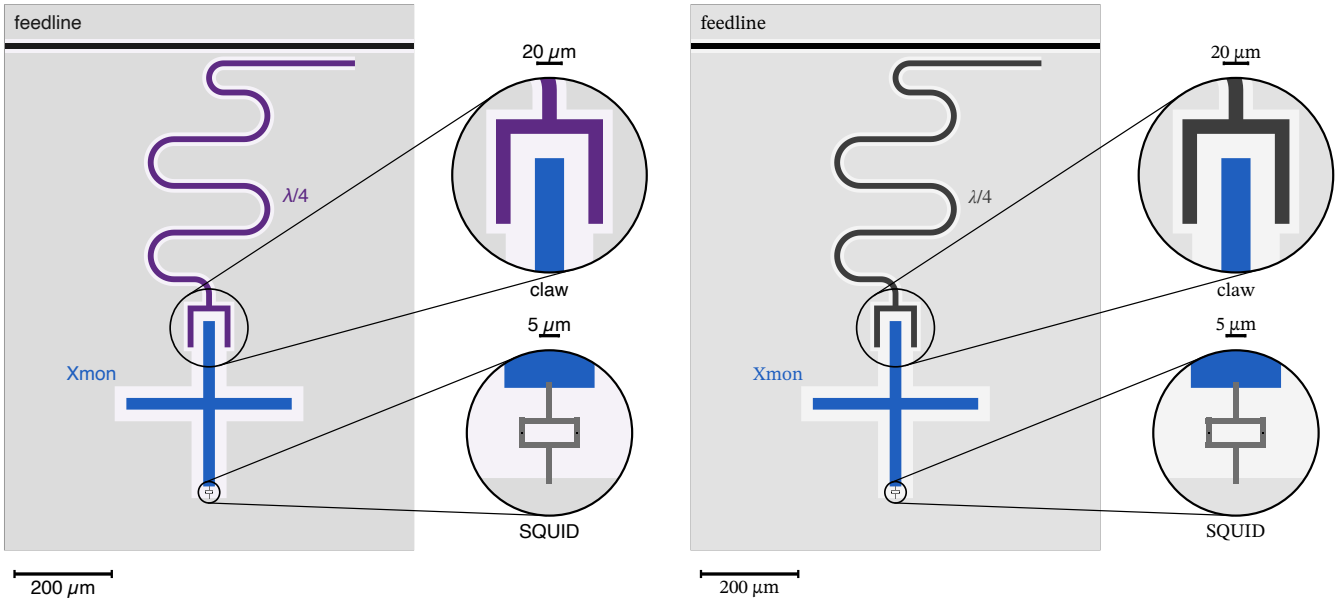
A generic literature-screening flow chart: reports pass a screening box, the exclusions branch off a trunk, a dotted arrow marks the citation search that adds reports to the full-text set, and the kept reports split into labelled categories (made-up counts).



86 Mask cell with magnifiers

`86_mask_zoom.py`

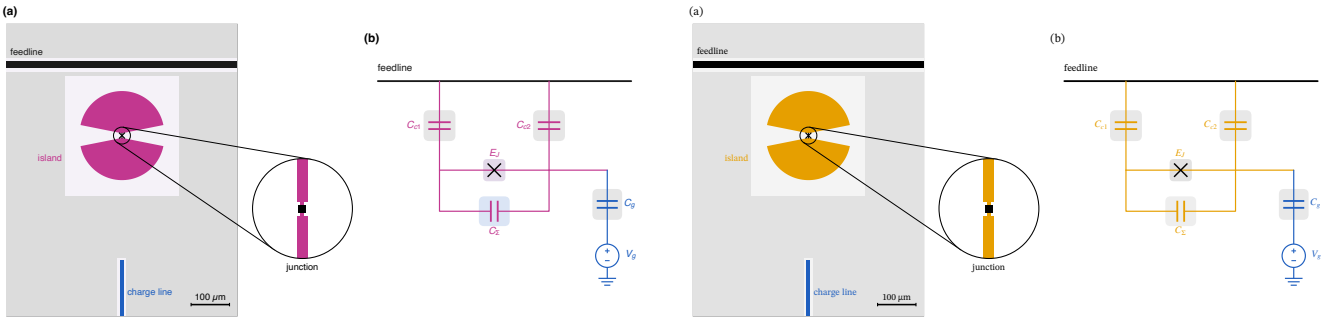
Xmon cell drawn to scale: a claw couples the quarter-wave resonator, a mask-level SQUID bridges the bottom gap, and two circular magnifiers show the claw and the SQUID, still to scale (designed geometry).



87 Bowtie sensor

`87_bowtie_sensor.py`

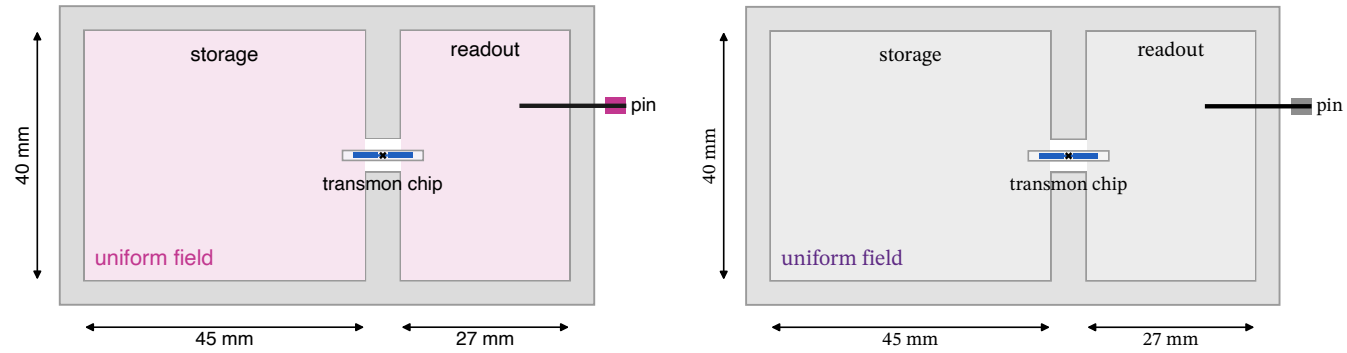
Resonator-free bowtie sensor: (a) the two wedge islands to scale beside a feedline, with a charge line below and a schematic magnifier on the junction; (b) its circuit, with the gate set by a dc source through C_g (designed geometry).



88 Cavity plan

`88_cavity_plan.py`

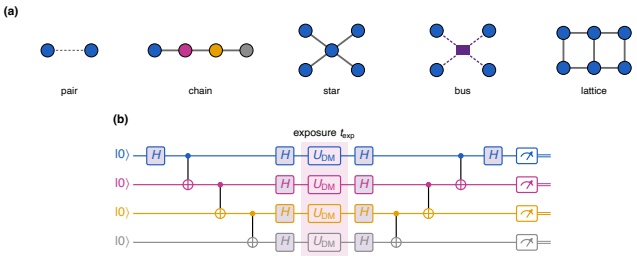
Two-chamber 3D cavity in plan view, to scale: a storage and a readout chamber joined by a trench that holds a transmon chip with two antenna pads, a coupling pin through the readout wall, and a uniform field washed over both chambers (designed geometry).

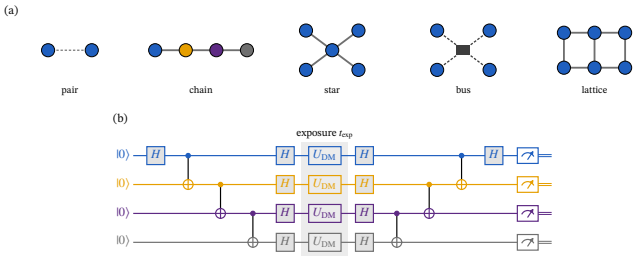


89 Topologies and a coloured circuit

`89_topology_protocol.py`

(a) Coupling graphs of five chip topologies: two qubits without a coupler, a chain, a star, a shared bus and a 2 x 3 lattice. (b) The four-qubit chain's sensing circuit, each wire in its qubit's colour: GHZ encoding, a shaded exposure window where a field acts on every wire at once, decoding and readout.



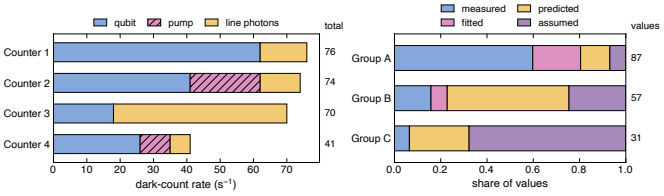


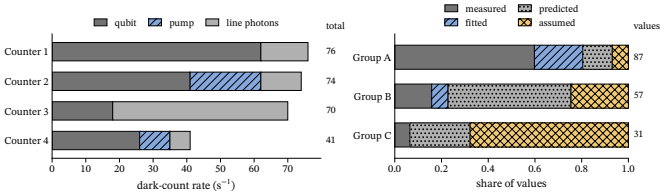
Comparison

90 Stacked budgets

`90_stacked_budget.py`

Dark-count budget of four photon counters split by source with each total printed beside its bar, and the share of four value classes in three study groups (synthetic numbers).

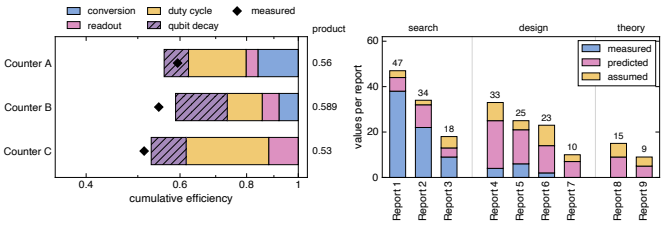


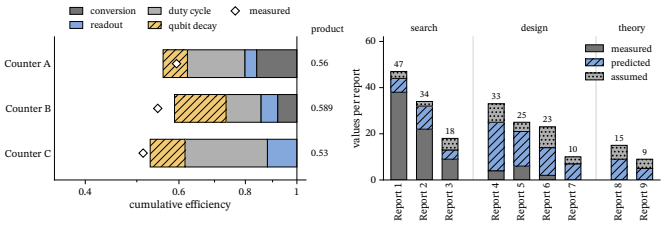


91 Efficiency chain

`91_efficiency_chain.py`

Detection efficiency of three photon counters as a product of four factors on a log axis with the measured efficiency marked, and parameter values per report stacked by status and grouped by study type (synthetic numbers).

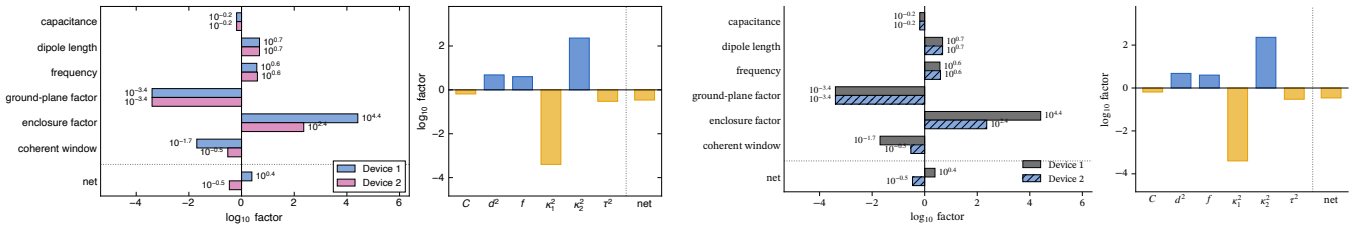




92 Factor budget

92_factor_bars.py

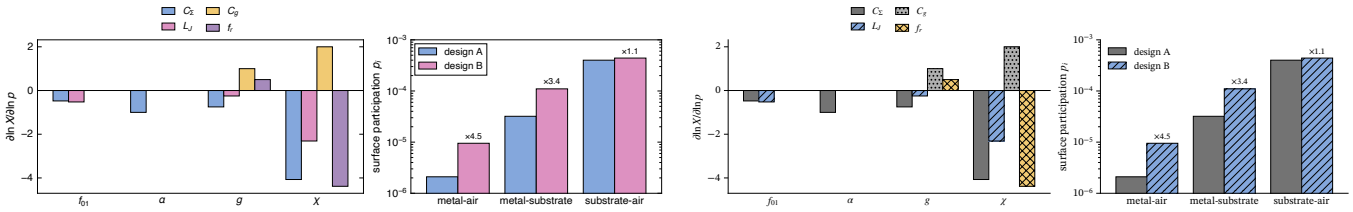
Signal of two devices against a reference design as signed log factors with the net change, and one device's budget as bars coloured by sign (synthetic numbers).



93 Grouped bars

93_grouped_bars.py

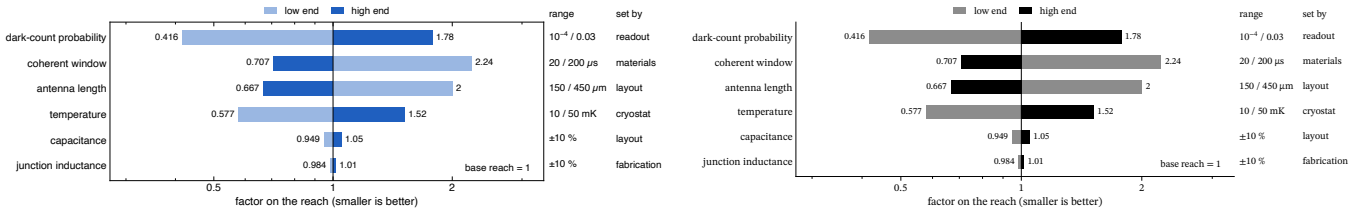
Logarithmic sensitivities of a transmon's frequency, anharmonicity, coupling and dispersive shift to four design parameters (computed), and surface participation of two designs with the change printed over each pair (synthetic numbers).



94 Tornado

94_tornado.py

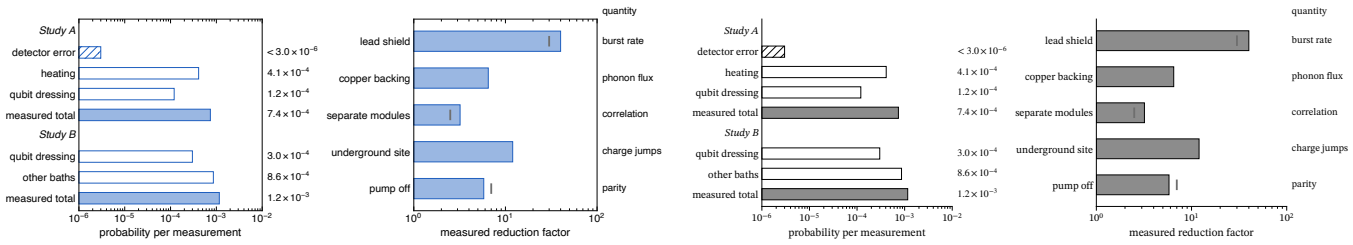
One-at-a-time sensitivity of a projected coupling reach to six design inputs, each moved to the ends of its range, with the range and the design step that sets each input (computed from a power-law model).



95 Bar rows

95_bar_rows.py

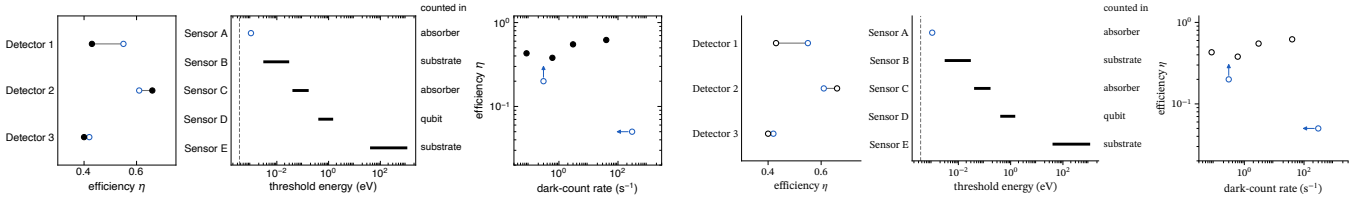
Background probability per measurement in two studies, with a hatched upper limit, open simulated terms and the values printed beside the bars, and measured reduction factors drawn out from 1 with published values as grey ticks (synthetic numbers).



96 Dot rows and bounds

96_dot_rows.py

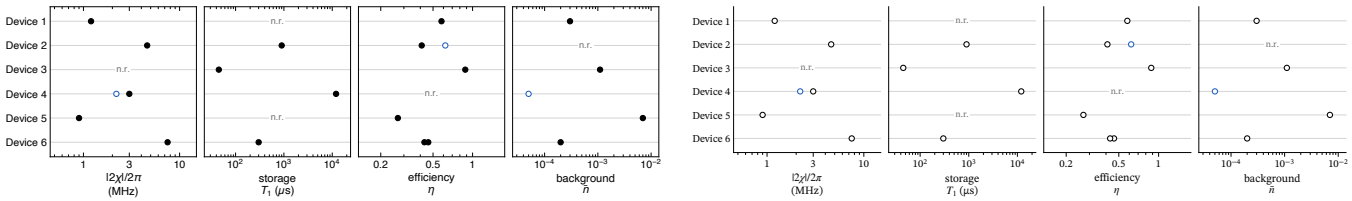
Predicted (open) against measured (filled) efficiency per detector, energy thresholds per sensor as points and ranges with where each energy is counted, and efficiency against dark-count rate with arrows on the two bounds (synthetic numbers).



97 Study strips

97_study_strips.py

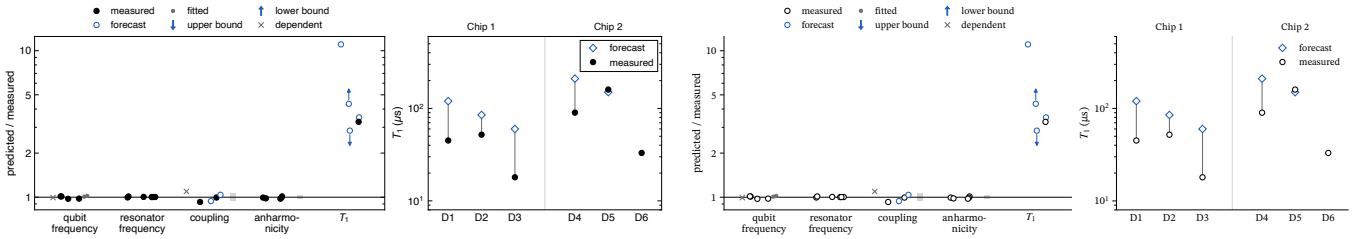
Four quantities across six devices on strips that share the device rows, each on its own log axis; filled markers are measured, open markers simulated, and n.r. marks a quantity a source does not report (synthetic numbers).



98 Ratio strip and stems

98_ratio_strip.py

Predicted over measured values by parameter type with each point's status and published scatters as grey bars, and forecast (open) against measured lifetime per device joined by a stem, grouped by chip (synthetic numbers).



99 Timeline

99_timeline.py

Reports by year in five research lines, one marker per report shaped by its kind; the sparse years before 2018 share a compressed stretch of the axis, and reports of one line and year sit side by side (synthetic record).

